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Birthplace Favoritism Revisited: Replication, Modern Evidence, and Crisis Dynamics

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Abstract

This paper extends previous work which shows that birthplace favoritism—the channeling of economic resources to a national leader’s home region—is a robust feature of global politics. We first verify their core findings from the earlier literature using an entirely open source framework combining leader birthplaces (PLAD) and subnational boundaries (GADM). We confirm the original findings and note that they are strengthened with extended temporal and geographic coverage. The analysis is extended into the modern satellite era (2012–2025) using NASA’s superior VIIRS based luminosity data. High temporal frequency of this dataset allows us to show that birthplace favoritism emerges immediately upon a leader’s accession and remains sticky after departure, contrasting with lagged and transient effects suggested by earlier lights data. We also discover a sudden and sustained birthplace premium coinciding with the onset of the COVID-19 pandemic. We show that this COVID bump is confounded by the systematically different characteristics of leader birthplaces, in particular by population density. We incorporate a two-pronged identification strategy to isolate the pandemic premium: (i) high dimensional country-month-population density fixed effects, (ii) using past leader producing regions as a more comparable control group. Both approaches yield a substantial and comparable surge in favoritism during the crisis. The pandemic premium is demonstrated to be concentrated in countries with weaker democratic institutions. These findings suggest a robust, contemporary evidence of birthplace favoritism that intensifies during crises, in particular for countries with weaker checks and balances.

1 Introduction

"The study of politics is the study of influence and the influential." Harold D. Lasswell
- *Politics: Who Gets What, When, How* (1936)

Does a national leader’s personal identity have an effect on the differences on the regional economic trajectories? This is a central question in distributive politics which focuses on how resources are allocated and whether and how political favoritism, i.e. skewed allocations based on subpopulation characteristics such as geography, ideology and ethnicity, plays a role in tipping the scales. A long list of anecdotal evidence, especially from countries with weaker institutions,

including Sri Lanka’s prime minister Rajapaksa, Libya’s Gaddafi, and Zaire’s Mobutu¹ suggests a pattern of diverting resources to their birthplaces.

One of the main hurdles in the analysis of distributive politics in a cross country scale is the inability to align measures of favoritism along with the added difficulty of doing this at a granular level. Golden and Min (2013) note that by the time of their review of research on distributive politics, this gave rise to research focusing on single countries. Moreover, they point out the uneven inclusion of research involving democratic countries because of availability and quality of data for the analysis. Hence, examination of systematic and global patterns of political bias has been hindered by such concerns.

The landmark study of Hodler and Raschky (2014), hereafter H&R, tackles the problems cited above by ingeniously incorporating new measures of highly granular economic activity into the realm of political favoritism. Henderson et al. (2012) provide the first attempt at proxying subnational economic activity by measuring nighttime light (NTL) intensity detected through the Defense Meteorological Satellite Program. H&R question whether nighttime lights are brighter for birthplaces of national leaders during their tenure. Using a specification with region and country-year fixed effects, they show a salient positive effect of being a leader birthplace on the NTL intensity in a panel of 126 countries from 1992 to 2009. This study provided a foundational benchmark by unlocking the power of Henderson et al. (2012)’s data innovation as well as reporting the first global scale evidence of the existence of regional favoritism.

In this paper, we reproduce and extend H&R in three main directions. Our first contribution is a replication of their main results in both narrow and wide senses² using alternative data sources in an open-source framework. Leader birthplaces and geospatial data indicating subnational boundaries used by H&R are proprietary. We argue that the ability to replicate key findings with publicly available datasets enhances the credibility of both the datasets and the original work, while providing a validated benchmark for subsequent extensions. Our initial inquiry confirms a broad compatibility with H&R’s main findings, particularly for lagged leader birthplace effects. Moreover, it shows that the results strengthen when extended temporally and geographically³.

Our second contribution asks whether regional favoritism persists into the post-2012 era where we incorporate a new generation of NTL data from NASA’s VIIRS sensors as part of the Black Marble project. Contemporary examples of national leaders potentially directing public resources toward their birthplaces still surface⁴. This includes criticism of the Nigerian president Muhammadu Buhari for favoring the ranchers of his home state Katsina over the rest of the country (Opejobi (2021)). In a similar vein, Mishra (2023) enumerates a list of public investments in Vadnagar, India’s prime minister Narendra Modi’s hometown—a region of population 28,000,

¹There are numerous examples of home town bias in allocating resources. For example, Beaumont (2011) in *The Guardian* lays down Muammar Gaddafi’s spending on his birth town of Sirte. Jones and Hameiri (2020) discuss China’s bailouts of failed megaprojects Mahinda Rajapaksa’s home town Hambantota. Mobutu Sese Seko spent millions of dollars building a palace and an airport in his home town of 1500 people, now in ruins (Smith (2015)).

²Journal of Applied Econometrics (n.d.) defines narrow replication as reproducing results of a scientific study broadly within the same data and context. Wide replications refer to potential extensions including temporal, spatial, or demographic coverage.

³Temporal extension refers to including 2013-2014 period for which additional DMSP-OLS data is available. We also consider spatial extension where we include the rest of the countries on top of the 126 countries H&R analyze

⁴These observations do not necessarily imply intentions of favoritism.

including plans to construct an airport.

DMSP-OLS datasets have been used extensively in economics research as their availability spans over 20 years from 1992 onwards. However, limitations of this NTL source has been discussed extensively, including coarse spatial and temporal resolution, lack of inter-sensor calibration, and top-coding issues. Even though NASA’s VIIRS sensors onboard the NPP and JPSS satellites provides solutions to these issues,⁵ Gibson et al. (2025), points to the slow adoption of VIIRS in economics. We address these concerns by providing an analysis that unlocks the capabilities of this new set of luminosity measures.

Our analysis using VIIRS NTLs confirms that the leader birthplace premium remains a robust feature of global politics after 2012. We find that the magnitude of the effect is comparable to DMSP-OLS period findings across a wide range of specifications. The estimates are generally stable and highly precise. They hold robustly for different sensor viewing angles, data quality filtering, and temporal aggregations of the monthly observations. Besides providing external validity of H&R’s framework, these results strengthen the case for using this new generation of NTLs in economics research.

Compared to DMSP-OLS’s yearly frequency, the higher (monthly in this case) temporal frequency of the VIIRS data allows us to investigate dynamic properties of favoritism with substantially more detail. To this end, we implement a high-frequency event study design to investigate luminosity dynamics around national leader changes. First, in contrast to H&R’s yearly DMSP-OLS framework that shows a lag, we find that the birthplace premium reflected by increased luminosity becomes apparent immediately upon the new leaders’ accession to power. Second, this effect has a *sticky* behavior in the sense that after a region ceases to become a leader region, there aren’t discernible differences in light emissions. This is also contrary to H&R’s analysis that is suggestive of an immediate drop to original levels after a following a leader’s departure from office.

Finally, encouraged by the precision of the results with VIIRS, we turn to the question of whether the economic benefits of birthplace favoritism trends are changing over time. To this end, we employ a specification that allows for time varying leader effect on a monthly basis. This dynamic analysis reveals a striking pattern: a sharp and sustained increase in the effect starting in early 2020, coinciding with the onset of the COVID-19 pandemic, and carrying on until late 2021. This observation, i.e. potential *COVID premium* to birthplace favoritism, is consistent with a growing empirical literature broadly considering political favoritism in aid allocation, and in particular focusing on COVID era policies.⁶

However, a core identification challenge towards interpreting this COVID premium as causal complicates the analysis. Regions producing leaders are not random, they are more populous, urban, and economically central⁷. Baseline COVID premium estimates would be confounded if these leader birthplaces are disproportionately affected by COVID shocks and accompanying

⁵A wide range of studies acknowledge the superiority of VIIRS over DMSP-OLS for economics research. See, for example, Shi et al. (2014), Gibson (2021), Gibson et al. (2021).

⁶Francken et al. (2012), Schneider and Kunze (2023), and Morales-Arilla (2024) provide instances of favoritism in the allocation of aid during natural disasters. Clemens and Veuger (2021), Matuszak et al. (2022), and Horwitz and Stephenson (2020) focus on skewed allocations during COVID-19, while Rose-Ackerman (2021) point out to the risks of corruption and unequal distribution of discretionary spending during COVID.

⁷H&R acknowledge that their leader regions has five times the population of non-leader regions.

government responses for reasons unrelated to being a leader region. Indeed, we confirm the scale of this concern with a placebo simulation exercise where we reassign leader status to other regions with probabilities proportional to their population density. These simulations trace the path of the birthplace leader effect remarkably similar to the actual trajectory, pointing out to the emergence of a spurious COVID-timed bump.

We address this challenge with a two-pronged identification strategy. First, building on H&R’s *country-year* fixed effects, we introduce *country-month-population density decile* fixed effects in order to flexibly absorb country specific shocks with respect to population density, a measure that purges shocks mostly related to urbanization. This strategy attenuates pandemic-era birthplace premium but estimates remain precise and economically meaningful.

To further strengthen inference, we construct a counterfactual based on a more comparable control group. We compare *COVID leader* regions that happened to produce the national leader during the pandemic with a carefully selected cohort of regions that produced *past leaders*. The understanding is that this latter set of regions share traits that make a region capable of producing leaders. An event study anchored to the start of the pandemic reveals no pre-existing trends, but an immediate divergence emerges starting in early 2020. These observations are confirmed by the application of the main specification to the regions that include only the regions that have ever produced national leaders. Finally, we cross validate the strength of our triple fixed effects approach above by exhibiting their effectiveness in wiping spurious leader premium due to being a past leader region.

Finally, we explore the role of institutional quality in curbing crises-induced favoritism, specifically by asking if democratic quality constrain a leader’s ability to divert resources in a crisis setting. Heterogeneity analysis involving polity scores indicate that COVID leader premium disappears for established democracies and are highly pronounced for the countries with weaker democratic quality.

Taken together, our analysis provides additional support and extended temporal coverage for birthplace favoritism as a global and still relevant phenomenon. Moreover, it intensifies under crisis conditions and is muted where democratic constraints are stronger. The remainder of the paper is as follows. Section 2 links the analysis to the existing literature. In section 3, sources and relevant details of the datasets are laid out followed by an analysis of the empirical strategy in section 4. Section 5.1 presents the results and the discussion, and section 6 concludes.

2 Related Literature

This section highlights the different strands of literature to which our analysis is related. As this analysis started as an exercise in replicating, we first provide a review of the understanding of replication in economics. Next, we move on to the themes of political/birthplace favoritism, including during crises and disasters, followed by a discussion of the use and analysis of NTLs.

2.1 Distributive Politics, Regional Favoritism, Economic Shocks

Distributive politics studies how governments allocate goods and services across population subgroups. Golden and Min (2013) distinguish programmatic allocations from patronage/pork

allocations where the former is concerned with allocations based on large subpopulations whereas the latter targets narrower subpopulations characterized by a set of shared characteristics between the policy maker and the recipient such as ethnicity and partisanship. They point out external validity and generalizability as persistent challenges: results often hinge on specific outcomes or countries. For example, studies including Burgess et al. (2015) observe ethnic favoritism in the specific context of road building projects in Kenya during non-democratic periods. On the other hand, Kramon and Posner (2013) caution that results regarding ethnic favoritism may not carry over to a different range of countries when other outcomes are considered. The landmark study of Hodler and Raschky (2014), which is the main focus of this paper, and De Luca et al. (2018) provide a neat solution to these concerns. They proxy regional GDP by DMSP-OLS captured nighttime lights and use it as the outcome in studies of birthplace and ethnic favoritism for large panels of countries; hence, focusing on a broad, welfare-relevant proxy for output.

Evidence of birthplace/hometown favoritism extends beyond national leaders. Elected or promoted agents in authoritarian regimes have been shown to favor their home regions in the context of China and Vietnam in the works of Mu and Zhang (2014) and Do et al. (2017). Moreover, footprints of birthplace favoritism by cabinet ministers (Widmer and Zurlinden (2022)) as well as national leader’s spouses (Bomprezzi et al. (2024)) is also studied. Mechanisms-wide, firms in leader regions enjoy better access to credit (Osei-Tutu and Weill (2024)), and external aid, specifically from China, flows to leader birthplaces (Dreher et al. (2019)).

Disasters, Economic Shocks, COVID-19 Pandemic

As Rose-Ackerman (2021) notes, it is no surprise that periods of crises, whether disasters or negative economic shocks, provide a fertile ground for heightened discretionary government spending decisions, that would have otherwise been subject to stricter scrutiny in normal times. The literature on crises relief suggests that political influence may shape the allocative outcomes. Directly extending H&R’s logic, Bommer et al. (2022) show that more US humanitarian aid is disbursed when a leader’s birth region is among the areas hit by a natural disaster. In a similar vein, Morales-Arilla (2024) note that Sub-Saharan African leader birth regions emit more night lights during drought conditions, and Ahlerup et al. (2024) emphasize how trust in government goes up in these regions during these periods and vice versa. Schneider and Kunze (2023)’s work points out the ambiguity channel as shown by how medium-level disasters present a chance for plausible deniability to mask partisan allocations.

COVID-19 pandemic is a case in point of unprecedented levels of discretionary government spending. Guasti and Bustikova (2022) provide an inventory of partisan irregularities in procurement and efforts to circumvent oversight in a subset of Eastern European countries, which they characterize as *pandemic heist*. Instances of political favoritism from Egypt in the form of government support for firms (El-Haddad and Zaki (2023)), and vaccine favoritism in Indonesia (Sari et al. (2023)) provide further support for this thesis. Moreover, Clemens and Veuger (2021) show that the political economy of pandemic favoritism is not only a feature of countries with weaker institutions by suggesting that political alignment with the federal government yields higher federal funding for the US states during the pandemic.

We extend this strand of the literature in three distinct dimensions. We first provide contemporary evidence of regional favoritism with higher quality data. Second, we contribute to

the growing literature on political behavior during crises by documenting a *COVID premium* to regional favoritism. This establishes the first global scale evidence of political favoritism during crises, specifically during COVID. Finally, our approach is able to address the moderating role of institutions for crises driven favoritism from a global perspective. We provide evidence that democratic institutions are highly effective in constraining exploitation of crisis for political gain.

2.2 Night Lights

Measuring development reliably at granular levels has been a challenge to which the seminal works of Henderson et al. (2012) and Chen and Nordhaus (2011) proposed DMSP-OLS nighttime lights as a solution, furthering the uptake of remote sensing data in economics (Donaldson and Storeygard (2016)). They used DMSP-OLS data, which is susceptible to blurring, top coding⁸ and calibration issues (Gibson (2021)), thereby creating mean reverting errors (Zhang et al. (2023)) that understate regional differences. Elvidge et al. (2013) highlight the superiority of the new generation VIIRS NTLs and Gibson et al. (2021) and Gibson et al. (2025) critically note the ongoing prevalence of relying on DMSP-OLS datasets in economics research resulting in underutilization of a valuable source of data. A case in point closely related to this study is that a significant number of research papers using VIIRS NTLs to investigate human behavior and several socioeconomic aspects of COVID-19 pandemic have been published in remote sensing journals⁹ whereas related exercises are scarce in economics journals¹⁰.

This paper makes two contributions to the literature that uses NTLs as a proxy for economic activity. First, it serves as a methodological bridge between DMSP-OLS and VIIRS datasets. Replication of H&R with DMSP-OLS and extension using NASA’s Black Marble suite provides an important cross-validation of this proxy. Our analysis strengthens the key findings in the literature based on DMSP-OLS data by providing evidence that the results are robust to using a dataset that doesn’t have the flaws of the original NTLs. Second, we provide a case study which takes full advantage of superior capabilities of the Black Marble suite. Our analysis utilizes the superior spatial and temporal resolution of the VIIRS sensors, as well as provides a comprehensive Black Marble workflow by incorporating different data layers like viewing angle and quality flags.

2.3 Replication in Economics

This paper is also in the category of extensions/scientific replications. We contribute to the replication literature in economics in two key regards. First, we perform a verification replication by reproducing the foundational findings of H&R’s influential study using entirely open-source datasets. This establishes a transparent and accessible benchmark for future research and addresses a novel scenario in replication taxonomies—validating results when the same underlying data is curated by different entities. Second, this study serves as a scientific replication and extension. Building upon our validated methodology, we uncover new dynamics of political

⁸Bluhm and Krause (2022) note that top coded pixels account for 32% of the entire light emissions.

⁹See, for example, Stokes and Román (2022), Shi et al. (2014), Alahmadi et al. (2021), Alahmadi et al. (2023), Dasgupta (2022), and Xu et al. (2021).

¹⁰An example is provided by Beyer et al. (2023) where they investigate the role of the severity of COVID-19 restrictions on NTL intensity.

favoritism that a simple verification exercise could not, demonstrating the value of replication as a launchpad for novel inquiry.

We replace the proprietary data of H&R with publicly available data sources and reproduce their core results—establishing an externally valid baseline on accessible datasets. This matches closest to a verification/pure replication even though the existing taxonomies do not fully address a situation where *same* data is curated by different entities. Our subsequent detailed tests of the birthplace favoritism hypotheses in the context of enhanced VIIRS NTLs and the COVID-19 period provides the core of our contribution, capturing a novelty above and beyond replication classifications noted above. This strategy mirrors a recent well regarded analysis by Kaplan et al. (2020) that first replicates a landmark paper of Mian et al. (2013) on accessible datasets followed by an extension that furthers the conclusions of the original paper. Replication studies in economics are rare. Mueller-Langer et al. (2019) report that only 0.1% of studies in the top 50 economics journals are replication studies despite their considerable disciplinary value. An explanation Galiani et al. (2017) provide is evidence of overturn bias, a form of publication bias where replications confirming original work being less likely to be published than the ones that don't. Moreover, Ankel-Peters et al. (2023) and Duvendack et al. (2017) point out the novelty bias, i.e. the influential journals' emphasis on novel contributions, as well as risks of replication exercises being perceived as adversarial in the profession, thereby contributing to the lack of such studies. Several taxonomies have been proposed regarding what replication entails—e.g. verification, reproduction, reanalysis, and extension (Clemens (2017)), or pure, statistical, and scientific replication (Hamermesh (2007)). Ankel-Peters et al. (2023) note that these definitions can be categorised based on whether the new work is identical to the original work with respect to the empirical specification, analyzed population, and drawn sample.

3 Data

3.1 Night-Time Lights: DMSP-OLS vs. VIIRS Black Marble

DMSP-OLS

Our primary measures of regional economic activity come from satellite-detected night-time light (NTL) intensity. For the period 1992–2013, we use data from the Defense Meteorological Satellite Program's Operational Linescan System (DMSP-OLS) collected by US Air Force Weather Agency, which was also the basis of H&R's original study. We obtain the annual "stable lights" composites for which image and data processed by the Earth Observation Group, Payne Institute for Public Policy, Colorado School of Mines. These composites report the average visible light intensity in a 30 arc-second pixel grid ($\approx 1\text{km} \times 1\text{km}$ at the equator) in digital numbers (DN) 0–63 over a year, filtering out ephemeral lights (e.g. fires) as well as discarding pixels tainted by sunlight, glare, moonlight, clouds, and aurora. We construct regional NTL measures by overlaying the DMSP raster on subnational polygons (see GADM below) and taking the average pixel value within each region-year. Following H&R, our dependent variable in replication is the logarithm of average night light intensity in a region-year.

Night lights from the *Operational Linescan System* (OLS) aboard DMSP satellites became a workhorse proxy for local economic activity after Henderson et al. (2012) argued that lights can

Table 1: DMSP–OLS stable-lights (key properties)

Property	Value / description
Availability	Annual composites, 1992–2013
Overpass time	~19:30 local time (evening)
Delivered grid	30'' (≈ 1 km) global raster
Quantisation	6-bit DN (0–63)
On-board averaging	5×5 IFOV block average (to save storage)
True footprint (per obs)	~ 2.2 km at nadir; grows off-nadir
Radiometric calibration	None recorded (nightly gain adjustments not stored)

inform growth analysis at *sub-and supranational* levels when conventional data are weak. By contrast, Chen and Nordhaus (2011) caution that DMSP measurement errors are substantial for both levels, further degraded by stable lights setting dimly lit low output regions to zero. Research on DMSP datasets points out to three core problems.

(i) *Overglow / blurring*. A significant "blurring" or "overglow" effect, where light from a bright source is spread out and wrongly attributed to neighboring, unlit areas, is observed in the DMSP datasets. Abrahams et al. (2018) identify three compounding sources. First, the *field of view is not constant*: while the nominal resolution of the DMSP sensors are 560x560 meters, the instantaneous footprint is nearly circular at nadir but becomes elliptical and much larger towards the edge of scan; creating an effective resolution of ≈ 2.2 km at nadir and exceeds 4.3km at the middle of the scan and 5.4km at the end of the scan (Elvidge (1997)).¹¹ Second, to conserve memory the satellite averages 5×5 samples on board raising the effective resolution to 2.8 kms. This positions a single light source across multiple pixel field of views and causes neighbouring pixels to see the same source¹². This causes both (i) same light source being counted multiple times in a 5x5 averaged pixel. (ii) towards the edge of each pixel neighboring pixels account for the same light sources, exacerbating the issue, now even further light sources are contaminating each other. Third, Tuttle et al. (2013)'s field experiments show rampant *geolocation errors*, light sources located on average 2.9 kms further away from their true geographic coordinates.

(ii) *Top-coding (saturation)*. A second important concern regarding DMSP-OLS composites is what is termed as top-coding where the upper range of the luminosity detections are truncated, resulting in a 6 bit DN's running from 0 to 63. Bluhm and Krause (2022) note that such truncation results in top coded pixels at about 2% levels across the world, corresponding to maxing out of DN's even for medium sized cities, impacting a wide range of world population as urban centers hold a disproportionate amount of people¹³. This creates discrepancies in two regards. First, averaging of 25 pixels in the 5x5 window contaminates top coded pixels with the others as long as there are pixels that has lower than 63 DN values. Second, as Hsu et al. (2015) notes, saturation effects are observed as early as at DN 35, and the severity is heavily pronounced by DN 55 according to Bluhm and Krause (2022), at which point they consider a pixel as top coded. These issues cause artificial flattening of bright areas, masking intra-urban

¹¹Gibson et al. (2020) note is the cutoff of data processing which mitigates elliptical expansion of the effective instantaneous field of view.

¹²Tuttle et al. (2013) observe that a single light source appears between 4-8 times in the 5x5 averaged pixels.

¹³For example, Gibson et al. (2021) show that based on 2012 dmsp values, 82% of Jakarta population lives in areas with 63 dn, and 17 with 62 dn.

variation and growth at the top.

(iii) *Lack of calibration (across time and satellites)*. OLS automatically adjusts gain (e.g., with lunar phase), but these settings are not recorded, so DN values are not radiometrically comparable over time or across overlapping platforms. Gibson et al. (2021) document large discrepancies across contemporaneous satellites, implying that time-series movements in DN can reflect gain as much as real economic change.

We use DMSP for historical replication and benchmarking, fully aware that its delivered 30" grid is not true 1 km resolution. Moreover, the aforementioned problems result in distortions in economic variables, as exemplified by Zhang et al. (2023) where the authors show that DMSP causes mean-reverting spatial errors and thus understates spatial inequality, an economic variable closely related to this research. Despite the shortcomings of DMSP NTLs, Gibson et al. (2020) observe that the switch from DMSP-OLS to VIIRS, which is detailed below, is slow in economics compared to other disciplines. By the date of their publication in 2020, DMSP was the forerunner in NTL space for Economics IDEAS/REPEC records. In this vein, we use DMSP primarily as a comparability bridge to the VIIRS era.

VIIRS Black Marble

We next turn to the new generation of NTL data from the Visible Infrared Imaging Radiometer Suite (VIIRS) to extend the analysis beyond 2013 as well as address the measurement issues related to DMSP-OLS data. The VIIRS sensor is jointly developed by NASA and NOAA and incorporated on Suomi National Polar-orbiting Partnership (Suomi-NPP, launched 2011) satellite and subsequent Joint Polar Satellite System satellites (JPSS-1/NOAA-20, launched 2017, and JPSS-2/NOAA-21, launched 2022). These satellites provide daily coverage with an equatorial overpass time of approximately 1:30 am. Table 2 summarizes key properties of the VIIRS datasets.

The Day-Night Band (DNB) sensor on VIIRS produces raw data that forms the basis of NASA's Black Marble product suite VNP46. VNP46A1 represents the raw at-sensor radiance data provided at 15 arc-seconds. The raw observations undergo a correction process to provide VNP46A2, which is the atmospherically corrected NTL radiance product. Román et al. (2018) detail these procedures, including correcting for atmospheric interference from water vapor and aerosol particles, removing stray light from lunar and solar illumination using a bidirectional reflectance distribution function (BRDF) correction, and taking into account ground reflectance changes due to seasonal factors like vegetation.

Monthly (VNP46A3) and annual (VNP46A4) composites are generated from the VNP46A2 data. Wang et al. (2022)'s compositing algorithm first removes lower quality observations due to clouds, aurora etc. A box metric outlier removal procedure is subsequently applied such that observations that fall outside of 1.5 times the interquartile range for each pixel are discarded.¹⁴ This effectively filters transient light sources for the upper end and possible power outages on the lower end. Pixels values that have no valid observations after the filtering procedure are *gap-filled* using data from historical observations. NASA provides quality flags regarding whether a pixel is gap-filled, and how many observations are used to derive the composite value.

¹⁴This procedure removes, for example between 1.5 and 6 % of yearly observations in Wang et al. (2022)'s test sites in Los Angeles.

¹⁵Version 1 of the data delivery format uses 16-bit quantization whereas Version 2 delivers the data in floating

Table 2: VIIRS Black Marble (key properties)

Property	Value / description
Availability	Monthly & annual composites, 2012–present
Overpass time	~01:30 local time (night)
Delivered grid	15'' (≈ 500 m) global raster
Quantisation	16-bit (representing floating-point radiance) ¹⁵
On-board averaging	None (full resolution data retained)
True footprint (per obs)	~742 m, constant across scan
Radiometric calibration	Yes (on-board solar diffuser)

The VIIRS sensor design and the Black Marble processing pipeline directly remedy the core deficiencies of DMSP-OLS. The VIIRS sensor design directly addresses the core deficiencies of DMSP-OLS.

1. *No blurring or overflow.* The patented detector design of the VIIRS provides a consistent instantaneous field of view which results in a constant ground footprint across the entire scan (Schueler et al. (2013)). This, combined with no averaging prior to processing, eliminates the blurring/overflow issues surrounding DMSP data. Furthermore, Wang et al. (2024) report that geolocation errors are negligible compared to DMSP-OLS.
2. *No top-coding.* The wide dynamic range of the sensor prevents top-coding, thereby allowing for accurate detection of light for even the brightest areas.
3. *On-board calibration.* VIIRS also incorporates an on-board solar diffuser which provides continuous radiometric calibration. As a result, the measurements are comparable across time and satellites.

The final data is delivered in a 15 arc-second grid (approximately 500m x 500m at the equator). The Black Marble product suite goes beyond providing a simple summary luminosity number. Each pixel has additional information attached and in particular, the following will be utilized in the rest of the analysis.

- **Snow-free observations:** Snowy conditions can substantially increase surface albedo—the amount of light reflected from the ground— which would inflate NTL readings artificially¹⁶. For example, Wang et al. (2022) note that Quebec City NTL values during snowy periods are double that of snow-free ones. Following these concerns, our primary analysis is based only on observations that are flagged as snow-free.
- **Pixel quality:** We use all available observations for our main analysis. On the other hand, we will take advantage of quality flags to conduct robustness checks that filters out the observations that are gap-filled or generated from a low number of daily observations.

point format. Floating point format retains extreme luminosity readings such as gas flares, which is not of interest to this paper.

¹⁶Wang et al. (2021) note that surface albedo is around 0.8 during snowy conditions compared to less than 0.2 in snow-free situations.

- **Viewing angles:** The Black Marble products are available with observations stratified by sensor viewing angles (zenith angle) (Wang et al. (2024)). Near-nadir angles (between 0–20° zenith angles) provide the clearest view, but they result in fewer observations (a median of 6 per month in a test site from Wang et al. (2022)). Off-nadir observations (between 40–60° zenith angles) provide more observations (10) at the cost of accuracy as longer distances cause higher atmospheric distortion. We test specifications using near-nadir, off-nadir, and all-angle composites, and our primary analysis will be based on near-nadir observations.

3.2 Leader Birthplace Data: PLAD

A key contribution that our analysis provides is using publicly available data to replicate and extend H&R’s results. In this regard, the original study relies on a proprietary leader birthplace dataset and instead we build upon the Political Leaders’ Affiliation Database (PLAD) provided by Bomprezzi et al. (2024). PLAD, as does H&R, extends the Archigos Database on Political Leaders which identifies the *effective* national leader for countries worldwide Goemans et al. (2009). An effective leader holds the ultimate executive power such as the presidents in presidential systems, the prime ministers in parliamentary systems, or the party chairperson in communist states. The version of PLAD we use covers the years 1989–2020 and provides information on 1,263 leader periods across 177 countries.

We enhanced the base PLAD data in two regards. First, we extended the temporal coverage from the end of 2020 through March 2025. This process added 145 new leader periods to the birthplace dataset. Second, we harmonized leader birthplaces with the subnational administrative boundaries (ADM2 level) to be used in the geospatial analysis. The original data lacked ADM2-level information or contained inconsistent ADM2 specification with our GADM map data for a large number of entries. We manually updated these entries, identifying ADM2 locations where possible. These updates are based on a variety of online results curated from sources such as encyclopaedic entries, official government websites, and news reports.

From this harmonized dataset, following H&R, we construct $Leader_{ict}$, a binary indicator equal to one if subnational region i in country c is the birthplace of the nation’s effective leader during time period t . How to treat partial period tenures, be it monthly or yearly data, presents a methodological problem. This is critical in the context of starting and ending periods of leader tenures, as well as in the case of brief leadership spells. The former remains a potential issue for all leader tenures. For example, for yearly data, it is questionable if a leader who takes office in December should be credited for influencing the entire year’s economic outcome. For the latter, the data suggests that 58 of the 1,412 total leader-periods in our dataset last for less than one month. As a judgment call, we decide on defining a region as a leader birthplace for a given period if the associated leader has been in power during at least a quarter of that period’s duration. Finally, in line with H&R, we exclude leaders born outside of countries in which they govern.

3.3 Administrative Boundaries: GADM

Our analysis requires matching NTLs to leader birthplaces, and to this end we need digital maps corresponding to subnational boundaries. In contrast to H&R, who uses data from the Center for International Earth Science Information Network (CIESIN) that is not publicly accessible to date, we utilize the open source Global Administrative Areas (GADM) database, version 4.1 GADM (2025).¹⁷

In accordance with H&R, we define our regions at the second administrative level (ADM2) where available. These administrative units are equivalent to districts and counties depending on the specific context of the considered countries. To maximize coverage, we apply the following harmonization procedure for the GADM shapefiles.

1. For countries where ADM2 levels are available in GADM, we directly use the provided polygons.
2. For countries that lack an ADM2 level in GADM, we upcast the first administrative level (ADM1) to serve as ADM2's. This creates a "pseudo-ADM2" layer, ensuring global coverage.¹⁸

H&R note that in their data, leader regions are 40% larger in area. For the GADM maps, we calculate that leader regions are on average 6% smaller across countries. This discrepancy is mainly due to differences in maps. H&R has 38427 total regions for their 126 country sample, and the corresponding number using GADM maps turns out 39,634 total regions for the same set of countries.

3.3.1 Population Data: GPWv4

One of the key issues that we identify for the rest of the analysis is related to disproportionate demographic characteristics of the leader regions compared to the rest of the relevant country. H&R note this pattern in their data, and we find a similar disparity in ours. In the global sample, we find that the mean population of leader regions are approximately 6.5 times that of non-leader regions, and the corresponding number for H&R's calculations in approximately 6. Moreover, the population density exhibits a ratio of 7.3:1 in favor of leader regions.

To get an estimate of population and population density numbers that will be useful for the analysis, we require granular population data at the subnational level. To this end, we turn to the Gridded Population of the World, Version 4 (GPWv4), Revision 11, produced by the Center for International Earth Science Information Network (CIESIN) at Columbia University (Center for International Earth Science Information Network (CIESIN) (2018)). GPWv4 provides a high

¹⁷The GADM project has a history of development since 2010s, though the current version of the project website provides limited information regarding its origins. Archived versions of the website from its earlier stages (e.g., version 2.8) show that GADM was developed by Robert Hijmans and collaborators at the University of California, Berkeley's Museum of Vertebrate Zoology and the International Rice Research Institute University of Berkeley, Museum of Vertebrate Zoology and the International Rice Research Institute (2012). We presume that the current version 4.1 is the latest iteration of this research effort.

¹⁸The following countries and territories fall into the category of regions for which ADM1 boundaries are used for ADM2-level analysis: Armenia, Bahrain, The Bahamas, Belize, Barbados, Comoros, Cabo Verde, Cyprus, Israel, Jamaica, Kuwait, Libya, Lesotho, Moldova, North Macedonia, Mauritius, Qatar, Singapore, Trinidad and Tobago, and Northern Cyprus. These regions typically correspond to small island nations or states with few subdivisions.

resolution raster dataset regarding the global distribution of the human population. The 30 arc-second grid (≈ 1 km at the equator) raster is based on population counts from census and administrative records. These population numbers are allocated uniformly to pixels within the corresponding administrative region boundaries¹⁹.

GPWv4 is a collection of different rasters regarding demographics and we, in particular, use the UN WPP-Adjusted Population Count raster for the year 2015²⁰. We pick this particular iteration mainly because 2015 is the available year closest to COVID-19 pandemic but did not interfere with the pandemic period. We next intersect this raster with our GADM adm2 polygons to compute zonal statistics with a specific focus on the population densities per each region. This will, in turn, allow us to construct within country population density deciles and quartiles, which will be the basis of the multi-dimensional fixed effects that will be employed to address identification issues.

3.3.2 Governance and Institutional Quality I: Polity5

Center for Systemic Peace (Marshall and Gurr (2020)) provides Polity5, which is an annual time series that characterizes different regime types for countries ($\geq 500,000$ population) from 1800–2018. The Polity project is a standard source for research on political regimes and in particular has been utilized in H&R as a proxy of institutional quality. Our analysis uses `polity2` index ranging from -10 (fully autocratic) to $+10$ (fully democratic), constructed as `democ - autoc`. Democ and autoc scores are created based on characteristics such as constraints on the executive and level of political competition.

Our empirical approach uses `polity2` index to explore whether the birthplace premium during the pandemic is mediated by institutional quality. To do so, we will focus on 2018 `polity2` scores as it is the final year of the data and is closest to the start of the pandemic. To test whether pandemic effects are concentrated in particular regime types, we create a variable *High-democracy*. This variable corresponds to strong democracies that is defined as the subset of countries that are roughly in the top tercile of the `polity2` scores. Out of 163 countries for which `polity2` scores are available, this corresponds to the top 70 countries for which `polity2` score is 8 or higher in 2018.

4 Empirical Strategy

Our empirical analysis will unfold in three stages in line with our main objectives. First, we employ an identical empirical specification to H&R’s original study in order to replicate their results with the new publicly available datasets for the DMSP-OLS era (1992–2013). Once we establish this benchmark, which also serves as a quality check on our data, we extend the same model to the VIIRS era (2012–2025) datasets with higher temporal, spatial and radiometric resolution. Higher temporal frequency of the VIIRS data allows us to zoom into the dynamics of birthplace favoritism. Finally, we attempt to establish whether a COVID premium exists with

¹⁹Center for International Earth Science Information Network (CIESIN) (2018) note that version 2 of GADM shapefiles are used for mapping purposes.

²⁰This iteration of GPW uses 2010 generation of censuses and population counts which correspond to data collected between 2005-2014.

respect to regional favoritism where we address identification challenges using population density based multidimensional fixed effects as well as focusing only on leader producing regions.

The main outcome variable throughout the analysis is the log-transformed average NTL intensity in a region, i.e.

$$\text{Light}_{ict} = \log(\text{NTL}_{ict} + 0.01)$$

where i indexes subnational regions, c countries, and t time periods (years for DMSP, months/years for VIIRS). A small constant 0.01 is added before log transformation to account for regions with zero luminosity, which, otherwise, would have been eliminated during the transformation.²¹ Moreover, our key explanatory variable is Leader_{ict} , which is a binary indicator corresponding to whether region I is the birthplace of the national leader of country c in period t .

4.1 Replication of the Original Study

Our first task is to estimate H&R’s baseline annual fixed effects specification to replicate their findings, i.e.

$$\text{Light}_{ict} = \alpha_i + \beta_{ct} + \gamma \text{Leader}_{ict} + \varepsilon_{ict} \quad (1)$$

where α_i corresponds to region fixed effects which absorb time-invariant regional characteristics such as geography, demographics, and culture. Country-year fixed effects, β_{ct} , control for country-wide shocks and trends (e.g., national GDP growth, policy changes, natural disaster). They also implicitly control for worldwide shocks and trends, which includes changes to satellites, sensors, and data collection methodology. γ is the coefficient of interest, which captures the leader birthplace favoritism effect and is expected to be positive under the assumption of regional favoritism. As H&R note, such a causal interpretation requires overcoming the identification challenge that regions that get more economically active and emit more NTL over time should not be more likely to be a leader region during that period. H&R address this concern by providing an event study where they include extensive dummies for before, during and after being leader regions to see if NTL emissions outside of the leadership period diverge from the rest of the country. Finally, standard error are clustered at the level of leadership periods, i.e. country-years for which the same person or group of people²² are effective leaders.

4.2 Main Analysis with High-Resolution VIIRS Data

In order to extend H&R study temporally to 2012–2025 period, we turn attention to NASA’s VNP46 suite which provides a wide range of NTL products, in particular at daily, monthly, and yearly temporal aggregations. The yearly VNP46A4 product undergoes the same procedure as above, whereas for the monthly panel based on VNP46A3 product, we employ country-month fixed effects instead of the previous country-year fixed effects. Monthly analysis, as opposed to yearly observations, avoids averaging away the true effect over the years for which leadership changes within the said years. On the other hand, since data is based on fewer observations, it is more prone to seasonality and noise.

²¹In doing so, we follow the approach of H&R where the authors note that this constant helps interpret the coefficients as semi-elasticities.

²²This is possible, especially for yearly data, when there are more than one national leader in a given period.

Monthly composites of the VIIRS data allow us to identify leadership changes precisely; hence, making it possible to trace out the dynamic effects around these transitions. Therefore, we employ an event study approach which would be useful in detecting pre-existing trends and dynamics for newly crowned leader regions, as well as providing a picture of regions that lose leadership status around the transitions.

To this end, we first construct cohort specific stacks around leader transitions such that each treated region in an event is paired with same country regions that are never treated within the window. In particular, we include events with complete windows and never-treated controls. Gain and loss stacks are constructed separately. Next we estimate the following two-way fixed effects event study

$$\text{Light}_{ie\tau} = \alpha_i + \beta_{e\tau} + \sum_{\tau \in \mathcal{T} \setminus \{\tau_0\}} \gamma_{\tau} (\mathbf{1}_{\tau} \times \text{Treated}_i) + \varepsilon_{ie\tau}, \quad (2)$$

where i indexes regions, e leader transition events, and τ corresponds to the period²³ relative to the event time. α_i are region fixed effects; and $\beta_{e\tau}$ are *event* $\times$ *relative period* fixed effects used to absorb cohort-specific shocks common to treated and controls at each τ . Treated_i is an indicator that equals one for the treated region within event e . We set the baseline to $\tau_0 = -2$ for reasons that will be clear in the succeeding analysis and report γ_{τ} for all τ . Standard error clustering is done at the event level.

In this setup, each γ_{τ} compares the treated region to its control pool within the same event stack. It is interpreted as the difference between treated and control at relative period τ versus the baseline τ_0 . Therefore, pre-treatment coefficients $\tau < 0$ provide a placebo test of parallel trends, and $\tau \geq 0$ trace out the dynamic response of this treatment-control differential to the changes in leadership.

4.3 Temporal Dynamics and the COVID-19 Premium

Monthly VIIRS composites allow us to perform sharper timing tests as they provide the precision needed for studying calendar-time heterogeneity. As noted before, annual aggregation curbs valuable information contained in the calendar-time profile such as jumps and breaks; thereby, reducing visibility of specific events in the analysis. In particular, we use this to ask whether birthplace favoritism is intensified during COVID-19 pandemic, a phenomenon we call as pandemic premium (to regional/birthplace favoritism). We formally test potential amplification of favoritism during this crisis where we define the COVID window as January 2020–December 2021 using the following specification:

$$\text{Light}_{ict} = \alpha_i + \beta_{ct} + \gamma_0 \text{Leader}_{ict} + \gamma_1 (\text{Leader}_{ict} \times \text{COVID}_t) + \varepsilon_{ict} \quad (3)$$

Here γ_0 represents average birthplace favoritism outside of the pandemic window, and γ_1 is the pandemic premium. A positive and significant γ_1 potentially suggests that birthplace favoritism intensified during this crisis.

²³We run the specification for both quarterly aggregation and pure monthly profiles. Quarterly specification is preferred as monthly noise is substantially mitigated.

In this setup, the identifying assumption is that, conditional on region and country-month fixed effects, absent any favoritism, birthplace regions and others in the same country would have seen similar changes to NTL during the pandemic period. We note two main threats to this assumption. First, pre-trends in the sense that if NTLs in leader birthplaces were already on a steeper trajectory before 2020, γ_1 could reflect this trend rather than pandemic premium. To this end, we conduct the following placebo test where we roll back the pandemic window to a pre-pandemic period:

$$\text{Light}_{ict} = \alpha_i + \beta_{ct} + \gamma_0 \text{Leader}_{ict} + \gamma_1^{\text{placebo}} (\text{Leader}_{ict} \times \text{Placebo COVID}_t) + \varepsilon_{ict} \quad (4)$$

where Placebo COVID_t is an indicator corresponding to January 2016–December 2017 period. Such an approach mimics the estimation procedure and potential differential pre-existing trends regarding COVID pandemic with the benefit of not being a COVID period and is expected to capture these pre-trends if there is one. Hence, we expect $\gamma_1^{\text{placebo}}$ to be statistically indistinguishable from 0.

More significantly, the second threat to identification arises from leader birthplaces potentially differentially affected by the pandemic for reasons other than favoritism. We have, for instance, already shown that leader birthplaces are more populous and more densely populated. Consequently, this would be a major concern if pandemic’s impact on economic activity (hence NTLs) correlates with systematic differences as exemplified by higher population density; thereby, leading γ_1 to conflate favoritism with other COVID related changes linked to differential systematic characteristics. For example, pandemic related shocks such as tourism collapse, lockdowns focused on urban-heavy service sector, or transportation disruptions were more prevalent in areas with higher population density. Besides, behavioral responses might play a similar function. A particularly interesting relevant observation is the possibility of higher incidence of insomnia during the pandemic, especially for more urban areas, (see Morin et al. (2021); Voitsidis et al. (2020)). As the satellite overpass time is around 1:30am, the increased nocturnal activity might lead to the detection of higher NTLs in urban areas that could be falsely construed as favoritism.

We affirm and remedy this threat in several stages. First, in order to visualize the calendar-time profile of the leader birthplace premium, we estimate and graph the coefficients from the following specification:

$$\text{Light}_{ict} = \alpha_i + \beta_{ct} + \sum_{m \in \mathcal{T}} \gamma_t (\text{Leader}_{icm} \times \mathbf{1}\{t = ym\}) + \varepsilon_{ict}. \quad (5)$$

where $\mathcal{T}$ corresponds to the set of months in the sample.²⁴ This is followed by an exercise with placebo simulations that preserve population density exposure while decoupling the real leadership mapping. This is done by randomly assigning *placebo leader* status with probability that is proportional to a region’s population density within the country. Re-estimating 5, if the simulations follow a similar trajectory to that of the original sample, it would validate our concern that a naive comparison of leader regions to all other regions would be confounded by

²⁴Note that as there is no separate Leader main effect, we are able to estimate all γ_m ’s without a base month. Furthermore, since a large number of estimated monthly coefficients exist from January 2012 to March 2025, we find it more convenient to present them graphically.

density-linked COVID exposure. Next subsection addresses remedies to the issues surrounding COVID confounders.

4.3.1 Methodological Approaches to a Solution

Strategy 1: High-Dimensional Fixed Effects

Our first approach acknowledges population density as the main confounding factor and attempts to control for this within the original framework of 3 by introducing country-month-density bin fixed effects, i.e.

$$\text{Light}_{ict} = \alpha_i + \beta_{ct \times d(i)} + \gamma_0 \text{Leader}_{ict} + \gamma_1 (\text{Leader}_{ict} \times \text{COVID}_t) + \varepsilon_{ict} \quad (6)$$

where the crucial addition $d(i)$ denotes the density bin (e.g. decile, quartile) of region i , and $\beta_{ct \times d(i)}$ are country-month-density-bin fixed effects. This approach non-parametrically absorbs the shocks common to regions in a density bin within the same country. Moreover, as will be detailed in the next section, it successfully neutralizes the spurious COVID premium introduced by the density-weighted placebo simulations above.

While its simplicity makes it an attractive option, density linked fixed effects come with several caveats. First, it is possible that pandemic premium partly operates through density correlated channels, which would be potentially absorbed by these fixed effects. This could be true, for instance, if the pandemic spending decisions that are borne out of prioritizing birthplace regions would masquerade as systematic benefits that target a wider density related measure. Second, and more importantly, population density is only one of the potential confounders. As leader producing regions would differ in other respects, density linked fixed effects is not guaranteed to capture other meaningful confounders.

Strategy 2: A More Comparable Control Group of “Past Leaders”.

We address the caveats of the first strategy by considering a cohort of *Past Leaders* as our control group. The core idea is that regions that have produced leaders in the past but not recently are a better counterfactual for the pandemic era leader regions because they are likely to share many observed and unobserved attributes (such as population density, demographics, education, political centrality etc.) that make a region capable of producing national leaders.

In particular, we consider Past Leaders as the regions

- that were leader birthplaces for at least 12 consecutive months in the past,
- whose last leadership term ended by December 2015,
- that have never regained leadership status after 2015.

We consider two implementations on the Past Leaders strategy. First, we estimate 3 on the restricted sample of COVID leaders vs. Past Leaders. Second, we further refine the analysis around a single event date, January 2020, which is the start of COVID-19 pandemic, and estimate the following dynamic model:

$$\text{Light}_{i\tau_q} = \alpha_i + \beta_{c\tau_q} + \sum_{\tau_q = -K, \tau_q \neq -1}^M \gamma_{\tau_q} (\text{COVID Leader}_i \times D_{\tau_q}) + \varepsilon_{i\tau_q} \quad (7)$$

where $\beta_{c\tau_q}$ are country-quarter fixed effects and $\text{COVID Leader}_i = 1$ for COVID leader birthplaces and 0 for Past Leaders. We are interested in the coefficients γ_{τ_q} which compare COVID leaders to Past Leaders at each τ_q relative to the baseline $\tau_q = -1$. The pre-event coefficients ($\tau_q < 0$) check parallel trends whereas post-event coefficients ($\tau_q > 0$) estimate the dynamics of the treatment effect. Standard errors are clustered at the country level.

Weaving the Two Strategies: A Two-Step Placebo Analysis

In this final step, we address two questions regarding our two empirical strategies by means of a two-step placebo simulation exercise. In the first stage, for each leadership term, we reassign the current region to a different region sampled from regions that produced a leader prior to 2012 within that country. As these leader-like regions are more similar to the leader regions before the swap, this is expected to create a significant, yet spurious, pandemic leader birthplace premium when applied to 3. Hence, such an observation will further bolster the arguments related to similarity of regions from COVID Leaders and Past Leaders.

Second, we will use our preferred empirical strategy of using density linked high-dimensional fixed effects to see how effective this strategy is in eliminating the spurious premium. To this end, we will apply equation 6 to the placebo leader samples produced above and measure how much of the pandemic premium is left after the introduction of the density bins. If the remaining premium is relatively small (or negligible in the optimistic scenario), this will give us confidence in using these density linked fixed effects for the estimates in the main analysis.

5 Results

This section will proceed in three main stages. We first begin by validating the empirical setup through replication of H&R’s baseline setting using DMSP-OLS NTLs where we only use open source datasets. The next step will be concerned with extending the analysis to post-2012 period using a state of the art VIIRS sensor based NTL suite made available by NASA’s VNP46 products. This subsection will feature an extensive set of applications highlighting the empirical value of the VIIRS based data due to its higher radiometric, spatial, and temporal resolution, as well as quality flags. These will allow us to implement precisely timed leader transition event-study designs. Next, we will probe calendar-time heterogeneity with a focus on the COVID-19 pandemic period to see if regional favoritism levels are heightened during this crisis period. A detailed two-pronged empirical strategy will be used to assess the positive findings. Finally, we investigate the role of democratic institutional quality in curbing the excesses during the crisis.

5.1 Replication and Extension with DMSP-OLS Data

We begin the empirical analysis by presenting the results from the replication and extension of H&R using DMSP-OLS NTLs. Our primary goal in this exercise is to verify H&R’s foundational findings when using our open-source data framework, which, in particular, relies on the publicly available PLAD leader database and GADM administrative boundaries. In turn, this validation provides a benchmark before we proceed to the VIIRS NTL analysis.

Table 3 details the results from the annual for the period for which DMSP-OLS data is available. We structure the analysis in three parts: a direct replication of the H&R sample and

Table 3: Replication and Extension of H&R with DMSP-OLS (annual panels)

Dependent variable: Light _{ict}	(1) Lag 0	(2) Lag 1	(3) Lag 2
Panel H&R: H&R Main Results (1992–2009)			
Leader _{ict}	0.038*** (0.014)		
Leader _{ict-1}		0.039*** (0.015)	
Leader _{ict-2}			0.041*** (0.013)
Panel A: Replication of H&R Sample (1992–2009)			
Leader _{ict}	0.027* (0.016)		
Leader _{ict-1}		0.042*** (0.016)	
Leader _{ict-2}			0.045*** (0.016)
Panel B: Temporal Extension (H&R Sample, 1992–2013)			
Leader _{ict}	0.033** (0.016)		
Leader _{ict-1}		0.047*** (0.016)	
Leader _{ict-2}			0.052*** (0.016)
Panel C: Full Global Sample (1992–2013)			
Leader _{ict}	0.036*** (0.013)		
Leader _{ict-1}		0.046*** (0.013)	
Leader _{ict-2}			0.049*** (0.013)
Observations (Panel H&R)	690,495	690,495	689,870
Regions (Panel H&R)	38,427	38,427	38,427
Observations (Panel A)	712,375	712,184	711,648
Regions (Panel A)	39,618	39,618	39,618
Observations (Panel B)	870,847	870,656	870,120
Regions (Panel B)	39,618	39,618	39,618
Observations (Panel C)	1,042,190	1,041,913	1,040,888
Regions (Panel C)	47,467	47,467	47,467
Region FE	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes

Notes: Table presents results from annual fixed-effects regressions. The dependent variable is the log of average nighttime light intensity plus 0.01. Panel H&R reports the results from H&R, Panel A restricts the sample to the 126 countries and years in H&R. Panel B extends the time period to 2013, the end of the DMSP-OLS data availability. Panel C uses full global sample. Standard errors, clustered by country-years where the same group of leaders are in office, are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

time period 1992–2009, a temporal extension of that sample through 2013, and a geographic extension to our full global sample. Each row and column of Table 3 corresponds to a unique estimation depending on the lag levels of the Leader variable as well as the temporal/spatial extent of data. We also provide Panel H&R in Table 3 for completeness.

Panel A reports the results for the replication exercise in the narrowest sense, restricting to the identical 126 countries and 1992–2009 sample window considered in H&R. The headline result of this subsection is the close alignment of our estimates with theirs. Specifications with one and two year lags of leader ascension are positive and highly statistically significant, with coefficients of 0.042 and 0.045 respectively. These are within 10% of the benchmark values reported in H&R. Hence, similar to H&R, our results imply that leader regions experience about 4–5 percent brighter lights in the years following gaining leader region status after accounting for region and country-year effects.

In contrast to H&R’s highly significant estimate (0.038), the contemporaneous coefficient is smaller (0.027) and only significant at 10% levels. We understand that such discrepancies are plausible for several reasons. First, we use GADM boundaries and they may not perfectly align with the CIESIN data used by H&R. Indeed, observation/region counts in Table 3 is indicative of this misalignment. The ADM2 levels produced by the maps we use are slightly higher in number (38,427 vs 39,618). Moreover, PLAD leader database (and our subsequent modifications) may have minor differences in leader term dates or birthplace locations. Finally, and potentially most consequentially, we consider a partial term a leader year for the corresponding region if the said leader was in power for a quarter of that period, 90 days in this case. How H&R tackle this technicality is unclear. Depending on whether H&R use higher or lower (or none) threshold, it may have a direct impact in the estimates regarding contemporaneous coefficients.

The next set of panels serve as wide replication exercises. In Panel B, we extend the DMSP-OLS dataset to include the rest of the years it is available, i.e. 1992–2013 period. On the other hand, we keep the same country sample as in H&R. The magnitude of the estimates increase and the standard error remain similar across the board, leading to stronger significance levels. The takeaway from this exercise is the suggestion that birthplace favoritism not only persisted, but also intensified towards the end of the DMSP-OLS data collection period.

Finally, in Panel C, we expand the analysis to include our full global sample, which includes numerous states and territories omitted from the original study, bringing the total number to 191. Appendix B lists these states and territories that are further included in our analysis. All three coefficients have lower standard errors and they are now highly statistically significant. The estimated magnitude of the leader effect remains stable and economically meaningful, reflecting a birthplace premium associated with a 4-5% increase in NTL intensity. This finding further strengthens the external validity of H&R’s framework, as another roughly 6% of the world population is incorporated into the analysis.

In summary, we accomplish two objectives in this replication exercise. First, we confirm the central finding of H&R. This is followed up by demonstrating that these results hold for wider temporal and geographic boundaries. These observations provide an anchor for what we will attempt to establish in the next steps: do these patterns of regional favoritism hold for superior NTL detections?

5.2 Regional Favoritism in the VIIRS Era

We now proceed towards the core of our analysis, extending the study of birthplace favoritism into the modern VIIRS era (2012–2025). Our main analysis uses the monthly (VNP46A3) and annual (VNP46A4) *near-nadir* composites of NASA’s Black Marble suite, which provide the clearest, least distorted view of ground-level light emissions.

Table 4 presents the baseline findings for the period from January 2012 to March 2025. The results are provided for both the global sample and, for comparisons to H&R, for the subset of countries that H&R analyze. All specifications utilize equation 1; hence, include region fixed effects and country-time fixed effects (country-month for monthly panels and country-year for annual panels). Moreover, standard errors are clustered at the level of country-periods for which the same leader or (potentially in case of leader transitions resulting in multiple leaders in the same period) group of leaders are in power.

Panels A and B of Table 4 presents results regarding estimates using monthly frequency with respect to lagged horizons for $k \in \{0, 6, 12, 24\}$ months. All lag structures in Panel A’s global sample yield a positive and precisely estimated birthplace premium. The coefficient estimates hover around 0.043–0.045, and interpreted as semi-elasticities, implying roughly 4.4–4.6% brighter leader regions, net of fixed effects. Panel B aligns the included countries with H&R’s 126 country subset. Magnitude of the coefficient estimates, still with high precision despite smaller sample, reveals around 5–5.4% higher luminosity during leader terms. The similar findings from Panels A and B suggest that global results are not an artifact of non-H&R countries.

We repeat the analysis using annually aggregated VIIRS datasets in panels C and D. This exercise establishes a direct comparison with the DMSP-OLS era analysis in terms of data frequency. The results closely corroborate those of monthly composites. Estimates corresponding to one- and two-year lags in these samples range between 5.1–6.1% whereas the contemporaneous effect is weaker at around 4–4.2% higher luminosity during leader periods.

Overall, analysis in this section finds that regional favoritism persists into the VIIRS era with magnitudes qualitatively and quantitatively similar to those found with DMSP-OLS. It is worth discussing several observations from this analysis. First, the coefficients in the monthly samples are highly stable across different lag structures. This stability suggests that the leader birthplace premium emerges fairly quickly after accession and persists at least two years after the leader steps down. Moreover, this carries over to the explanation regarding the disparity between contemporaneous and lagged values. Annual aggregation dilutes the effect of a leader taking office during mid-year, especially when the birthplace premium quickly takes effect. These observations highlight the value of having NTLs at higher frequency and this will also allow us to discuss leadership transition dynamics in the next subsection.

Next, we delve deeper into the comparison of DMSP-OLS and VIIRS analysis in the context of the years 2012–2013 for which both sensors offer observations. Table 4 shows a stronger relationship between VIIRS NTLs and birthplace premium compared to DMSP-OLS. One explanation, which is difficult to assess, is the possibility that this premium strengthened in recent years. Another explanation is related to how these different sensors operate. In Table 10 in the appendix, we include a log – log comparison between DMSP-OLS and VIIRS readings for various sample restrictions, specifically taking into account the fact that DMSP-OLS saturates

at 63, and as Bluhm and Krause (2022) note, a cell is likely top-coded by digital number 55. Moreover, we also analyze situations excluding light detections were not recorded for given areas. The main takeaway is that, the elasticity between DMSP-OLS and VIIRS is very close to 1, and slightly higher for when top (and bottom) coding is not a concern. Figure 7 provides a graphic illustration of this elasticity, as well as the extent of top coding. In this setup, it is therefore reasonable to expect that the birthplace premium readings are higher with VIIRS because for the denser regions that are close to being top coded from where leaders are more likely to originate, it is more difficult to detect differences before and after a leader takes office, attenuating the effect size. Finally, following H&R’s back of the envelope calculations regarding GDP-NTL relationship based on the elasticity of 0.3 suggested by Henderson et al. (2012), yearly data suggests a 2% birthplace premium over the regional GDP for the H&R sample of countries, and 1.5% for the full global sample.

5.2.1 The Dynamics of Leadership Transitions: A Matched Event Study

The panel estimates document a positive average effect size for the birthplace premium, but they do not detail when the gains are realized, or whether they subside. The previous analysis hints at what might be happening around leader transitions to be able to get estimates as given in the previous tables. However, this suggestion is at odds with the analysis put forth by H&R in their Figure III that uses the yearly DMSP-OLS data. The authors incorporate an extensive set of leader fixed effects into their main panel based on leader transition timing for before, during, and post leadership periods and argue that the leader gains are not realized until the third year, and drop immediately once the region loses leader status. As yearly analysis due to low frequency of the DMSP-OLS data allows only limited introspection, caveats of such an exercise due to bundling of leader periods that have highly different characteristics and durations ensue. To further investigate transition dynamics, we move beyond static fixed effect specifications and implement an event study that utilizes the monthly frequency of the VIIRS panel, allowing us to trace out the path of luminosity around exact periods of regions gaining or losing leader status.

The design features of this stacked event study is detailed in the Empirical Strategy section, built on Eq. 2. We construct the gain and loss stacks based on “clean” transition episodes. Before taking office, a leader region is required not to have had a leader status in the 4 quarters prior, as well as continuing the leader status 8 quarters after taking office. Similarly, for losses, a region is required to be a leader region continuously in the last 4 quarters and not regain leader status in the next 8 quarters. These are then matched with never treated regions from the same country-periods to finalize the stacks. The quarterly (monthly) coefficients, measured as the difference between the treatment and control groups with respect to its difference in $\tau_0 = -2$ quarters ($\tau_0 = -5$ months) are plotted in Figure 1.

The graphs show relatively flat pretrends for $\tau < 0$ across the board. There is a slight dip within the last period before leader transition. This may be partially be attributable to election spending etc. thus, we use two quarters (or 5 months) prior as the baseline period. Moreover, the post-transition dynamic picture shows a persistent pattern for both cases. Gaining leader status ($\tau \geq 0$) turns the treated-control gap positive quickly. The difference appears economically meaningful and propagates over the subsequent periods. By contrast, transition into a non-leader

Table 4: VIIRS Results: Full Sample vs. H&R Country Sample (2012–2025)**Panel A: Monthly VIIRS (VNP46A3), Full Global Sample**

	(1) Lag 0	(2) Lag 6	(3) Lag 12	(4) Lag 24
Leader _{ict-k}	0.043*** (0.013)	0.045*** (0.014)	0.045*** (0.013)	0.045*** (0.014)
Observations	7,254,532	7,254,532	7,254,532	7,253,993
Regions	47,517	47,517	47,517	47,517

Panel B: Monthly VIIRS (VNP46A3), H&R Country Sample

	(1) Lag 0	(2) Lag 6	(3) Lag 12	(4) Lag 24
Leader _{ict-k}	0.049*** (0.016)	0.052*** (0.017)	0.053*** (0.017)	0.052*** (0.017)
Observations	6,050,317	6,050,317	6,050,317	6,050,317
Regions	39,630	39,630	39,630	39,630

Panel C: Annual VIIRS (VNP46A4), Full Global Sample

	(1) Lag 0	(2) Lag 1	(3) Lag 2
Leader _{ict-k}	0.039** (0.016)	0.051*** (0.017)	0.050*** (0.017)
Observations	562,346	562,346	562,301
Regions	47,517	47,517	47,517

Panel D: Annual VIIRS (VNP46A4), H&R Country Sample

	(1) Lag 0	(2) Lag 1	(3) Lag 2
Leader _{ict-k}	0.041** (0.020)	0.058*** (0.022)	0.059*** (0.022)
Observations	471,986	471,986	471,986
Regions	39,630	39,630	39,630

Notes: Coefficients from the main specification in Eq. 1 for the VIIRS near-nadir products (VNP46A3 monthly; VNP46A4 annual), 2012–2025. The dependent variable is the log of average nighttime light intensity plus 0.01. Monthly panels include country-month FE; annual panels include country-year FE; all include region FE. Standard errors clustered by leader-terms (country-year/month cells sharing the same group of effective leaders) in parentheses. Significance: * 10%, ** 5%, *** 1%.

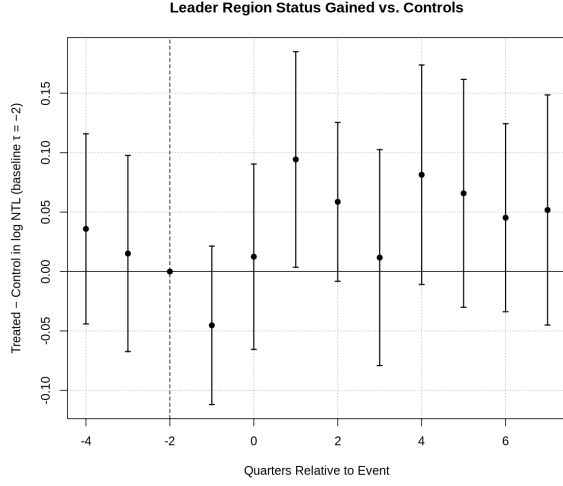
status doesn't exhibit analogous negative swings, NTL differential remaining relatively stable in subsequent years.

To recap, the event-time evidence provides a consistent explanation for the disparity between contemporaneous and lagged estimates of the birthplace premium. Contemporaneous estimates treat partial ascension years as if they are complete leader years and since we observe the premium fairly quickly after ascension, these estimates will discount the first year's premium. Furthermore, the one- and two-year lagged specifications appear similar in magnitude and larger than the contemporaneous estimate for two reasons. First, even though the first partial ascension year premium contributes to the untreated years' premium, potentially attenuating overall effect. On the other hand, this attenuation is highly diluted as the median leader tenures lasts around 2.5-3 years and the panel horizon is about 13 years. Second, the lagged values for the years that are coded as leader regions but correspond to the years right after the leader leaves but still positive due to the lagging procedure, are not impacted by the leader to non-leader transition. This is because the figures show that the premium differential stays stable around at least 2 years after the transition. These observations are also supported by the monthly results of Table 4 where the higher frequency tracks the premium more closely.

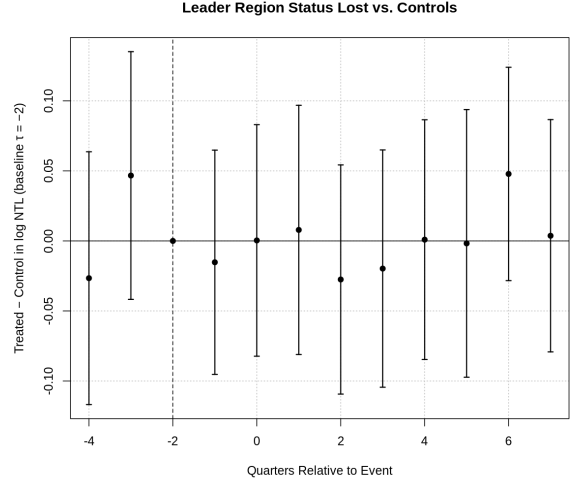
5.3 Robustness and Specification Checks

Our next task is to use Table 5 to assess whether NASA's Black Marble product suite based estimates are an artifact of the following breakdown.

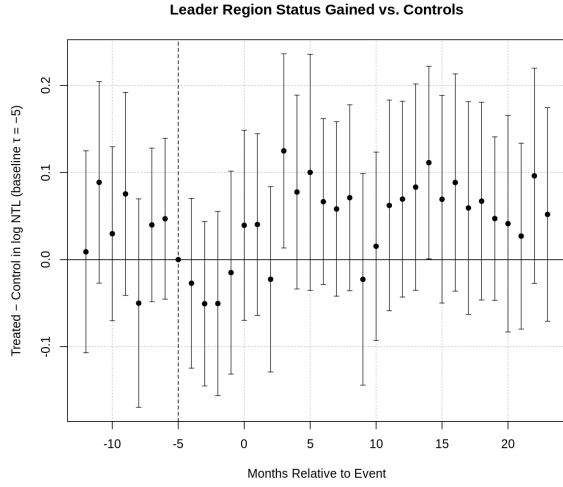
- i *product delivery at particular frequency*: NASA provides composites in daily raw (VNP46A1), daily corrected (VNP46A2), monthly (VNP46A3), and yearly (VNP46A4) formats. Monthly and yearly datasets are composited from the daily corrected observations. For the purposes of the analysis, we only use these two composites.
- ii *viewing geometry*: Depending on the zenith angle, ground observations within $0 - 20^\circ$ fall under near-nadir category, whereas $40 - 60^\circ$ viewing angles correspond to off-nadir observations. A third category, *all-angles* encompass observations regardless of the viewing angle.
- iii *temporal aggregation*: It is also possible to create different composites depending on the setup, and we find it to be potentially useful to provide quarterly, semi-annual, and annual averages of the monthly data.
- iv *masking based on observation quality*: Black Marble suite attaches a quality flag to each individual pixel for the monthly/yearly composites. This represents whether a pixel has *good-quality* or *poor-quality*, i.e. the number of daily observations used for compositing is greater than 3 or not, as well as whether it is calculated based on historical data on the said pixel. For Panels A and B of Table Table 5, we repeat the estimations by masking good-quality pixels. The procedure involves filtering out the pixels that have either poor-quality or historical data and running zonal statistics on the ADM2 polygons to calculate the mean based only on the good-quality pixels.



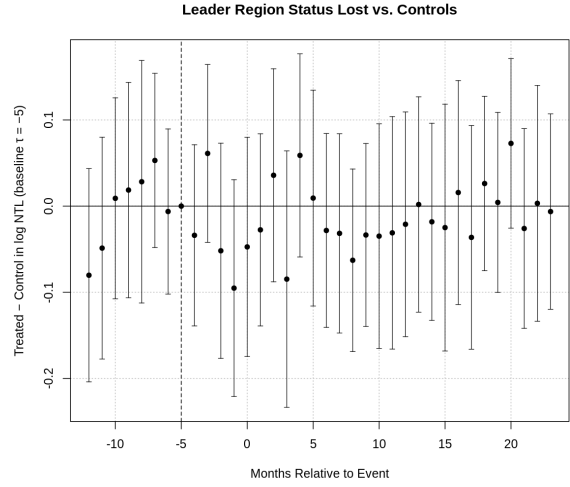
(a) Status gained (quarterly)



(b) Status lost (quarterly)



(c) Status gained (monthly)



(d) Status lost (monthly)

Figure 1: Dynamic effects around leader transitions (VIIRS, monthly basis).

Notes: Points plot event-time coefficients $\hat{\beta}_\tau$ from the matched difference-in-differences model in Eq. 2 with region fixed effects and event $\times$ relative-time fixed effects. For each “gain” or “loss” event, controls are never-treated regions from the same country and cohort window. The x -axis shows quarters (top row) or months (bottom row) relative to the transition; the baseline is two quarters ($\tau = -2$) for quarterly profiles and five months ($\tau = -5$) for monthly profiles. Vertical bars are 95% confidence intervals. The outcome is $\log(\text{NTL} + 0.01)$ from VIIRS (VNP46A3).

Table 5: Robustness Checks for VIIRS-based Estimates

		Lag (in corresponding number of years for yearly composites, months for all others)			
Data Product	Specification	0	6/1	12/2	24
Panel A: Monthly Composites (VNP46A3)					
All Angles (sublayer 5)	Standard	0.039*** (0.013)	0.040*** (0.014)	0.039*** (0.014)	0.035** (0.014)
	Masked	0.037*** (0.013)	0.039*** (0.014)	0.038*** (0.014)	0.035*** (0.014)
Near-Nadir (sublayer 15)	Standard	0.044*** (0.013)	0.045*** (0.014)	0.045*** (0.013)	0.045*** (0.014)
	Masked	0.036*** (0.013)	0.037*** (0.014)	0.041*** (0.014)	0.046*** (0.014)
Off-Nadir (sublayer 23)	Standard	0.034*** (0.011)	0.036*** (0.012)	0.030** (0.012)	0.025** (0.012)
	Masked	0.028** (0.011)	0.031*** (0.012)	0.029** (0.012)	0.025** (0.012)
Panel B: Annual Composites (VNP46A4)					
All Angles (sublayer 5)	Standard	0.034** (0.016)	0.043*** (0.016)	0.045*** (0.016)	
	Masked	0.035** (0.016)	0.043** (0.016)	0.045*** (0.016)	
Near-Nadir (sublayer 15)	Standard	0.039** (0.016)	0.051*** (0.017)	0.050*** (0.017)	
	Masked	0.039** (0.016)	0.051*** (0.017)	0.049*** (0.017)	
Off-Nadir (sublayer 23)	Standard	0.039** (0.016)	0.040** (0.017)	0.033* (0.017)	
	Masked	0.039** (0.016)	0.039** (0.017)	0.033* (0.017)	
Panel C: Aggregated Monthly Data (VNP46A3, Near-Nadir)					
Quarterly		0.043*** (0.013)	0.039*** (0.013)	0.038*** (0.013)	0.043*** (0.013)
Semi-Annual		0.034** (0.013)	0.032** (0.013)	0.031** (0.013)	0.035*** (0.013)
Annual		0.033*** (0.012)	0.035*** (0.012)	0.037*** (0.012)	

Notes: The table presents coefficients from the main specification in Eq. 1 using different Black Marble products and temporal aggregations. All models include region and country-time fixed effects. Standard errors clustered by leader-terms (country-year/month cells sharing the same group of effective leaders) in parentheses. Significance: * 10%, ** 5%, *** 1%.

The core result across all choices is the consistent message that leader birthplaces are systematically brighter than other regions in the same country during the leader’s tenure, with magnitude estimates ranging from 3 to 5 percent at conventional (and typically very strong) significance levels. Hence, this suggests that our analysis in the preceding (and succeeding) sections are not driven by our preferred near-nadir specification.

We first examine the effect of sensor viewing angles on our results. Wang et al. (2022) provide an instance of a $10^\circ \times 10^\circ$ tile area²⁵ where authors illustrate that the median number of number of observations is around 5-6 for near-nadir composites, 9 for off-nadir, and 24 for all angles for the pixels included in the tile during June 2016. Yearly observations exhibit a parallel structure. The results from both monthly and yearly panels (A&B) of Table 5 yield two main patterns. First,

Second, and more importantly, near-nadir angles provide coefficients that are largest and most precise regardless of the lag structure. Wang et al. (2022) note that while the relative magnitude of radiance levels with respect to the viewing angles depends on the characteristics of the testing site, near-nadir composites provide largest temporal variation among different viewing categories. On the one hand, we have a lower number of observations for the composites using near-nadir angles, which results in noisier observations. On the other hand, even though VIIRS solves the distortions regarding field of views compared to DMSP-OLS, its readings are still susceptible to capturing stray light originating from neighboring areas that is reflected off of the area in consideration. In this sense, wider viewing angles results in a degree of higher spatial averaging, potentially attenuating the inherent signal due to changes in leadership status of a region. Our analysis suggests that near-nadir viewing angles maximize the signal to noise ratio in our setup justifying the use of these composites for the rest of the analysis. This is also strengthened by the observation that yearly observations discount contemporaneous effect (which is expected to be the case because of partial-period tenures) for near-nadir profiles, but not for off-nadir profiles.

Finally, in Panel C, we aggregate the monthly near-nadir composites to quarterly, semi-annual, and annual averages. This is an exercise that would be useful in applications requiring different frequencies and interpretations. Besides, annual aggregation of the monthly composites is critically different from annual aggregation of the the daily observations. First, outlier removal procedure is applied once over complete set of observations in a year for the yearly composite whereas for the annual aggregation of monthly composites, this procedure is pre-applied to all monthly components. Second, and more substantively, the yearly composites give equal weight to all included daily observations. However, monthly composition of these observations is likely to be different because of quality filtering, most notably for weather events. For instance, as we are working primarily with snow-free observations, this will reduce the number of winter observations compared to the rest. Similarly, cloud cover is an important determinant of whether an observation would be included, which similarly is impacted differentially by seasonal patterns. Hence, the annual aggregation of monthly observations implicitly reweight the observations in

²⁵As the Black Marble products are extremely detailed; thus resulting in large file sizes, NASA delivers these products in $10^\circ \times 10^\circ$ tile sizes. We first select necessary bands, and then stitch together these tiles to get a geotiff file of the whole globe, which is then used for calculating zonal statistics such as the mean of NTLs per each ADM2.

the composites. As expected, the higher the frequency, the closer the estimates track the monthly counterparts. The contemporaneous dilution observed in the yearly composite is less pronounced in the annual counterpart of the temporal aggregation. On the other hand, the pattern is stable such that effect sizes of 3-4.5% are similar to what is observed in the monthly setup.

5.4 The Temporal Dynamics of Favoritism and the COVID-19 Premium

The high monthly frequency of the VIIRS data allows us to track the calendar-time profile of the birthplace premium. Plotting this monthly profile, we recognize a COVID-era shift (during January 2020–December 2021)²⁶ in favoritism. As we suspect that the COVID premium can be an artifact of a special interaction between leader regions’ characteristics and the systematic pandemic changes, we introduce placebo studies that clearly point to the role of population density as an important confounder. We next introduce two design choices to tackle concerns related to the identification of a potential COVID premium:

1. high-dimensional fixed effects to directly absorb country $\times$ month $\times$ density-bin shocks,
2. restricting the analysis to a control cohort of *Past Leaders* that are presumably similar to current leader producing regions.

We then introduce another placebo simulation based on Past Leaders and the high-dimensional fixed effects approach to validate our solution. The results section is finalized with a short study on how democracy levels summarized by polity scores mitigate the COVID premium.

5.4.1 Calendar-Time Path of Leader Premium

We follow Eq. 5 from Section 4 to investigate the calendar-time profile of the leader premium. In this setup, Leader indicator is interacted with a full set of year-month dummies, allowing the leader premium to vary freely over time. Figure 2 plots the resulting series of monthly coefficients. The visual calendar-time profile is striking. The monthly estimates heavily fluctuate around what it appears to be a positive mean prior to 2020. By the end of 2019, there is a sharp, sustained increase in the estimated monthly leader premium, coinciding with the onset of the COVID-19 pandemic. The estimates revert back to pre-pandemic levels as we move towards the end of 2021, a period where most pandemic policies and restrictions were lifted.

To formally quantify the pattern of heightened NTLs for leader regions during COVID-19 period, we estimate the model in Eq. 3 that isolates the pandemic premium. Column (1) of Table 6 presents the result from this specification where the coefficient on the interaction term, $\text{Leader}_{ict} \times \text{COVID}_t$, is large, positive, and highly statistically significant. The baseline favoritism coefficient is slightly lower than the version without the COVID interaction, and the pandemic premium is 4 times as large as the previous baseline calculations.

As discussed in the Empirical Strategy section, the pandemic premium measured above is rather a correlation and identification for a genuine pandemic premium requires that after accounting for country-month and region characteristics, birthplace regions and others in the same

²⁶We suggest this window as the particular COVID window because restrictions and lockdowns were lifted after this period for most of the globe.

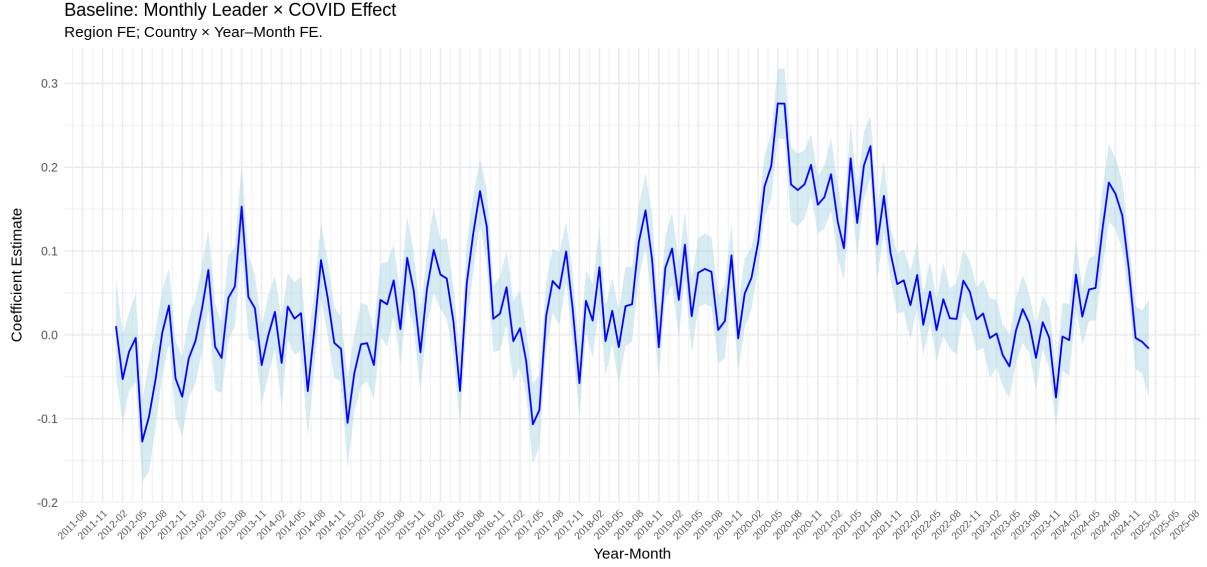


Figure 2: Calendar-time dynamics of the birthplace premium (VIIRS monthly, 2012–2025).

Notes: Points plot month-by-month coefficients from Eq. 5. There is no separate *Leader* effect; therefore, each coefficient represents how much the luminosity in a leader region is for each month. Coefficients are shown with 95% confidence intervals; standard errors are clustered at the leader-term level (country-month cells sharing the same set of leaders).

country would have been on parallel luminosity trajectories if birthplace regions were stripped of their leader status. We identify two major identification challenges towards uncovering such a pandemic premium. The first issue is whether leader premium would have already started to diverge prior to the pandemic, and the pandemic premium would have picked up this trend as well. To test the extent of this concern, we employ the specification from Eq. 4. This introduces the interaction term $\text{Leader}_{ict} \times \text{Placebo COVID}_t$. Placebo COVID_t is calculated by rolling the COVID period to four years prior. Second column of Table 6 shows that the exercise of rolling the pandemic period results in an insignificant placebo pandemic premium with a negligible magnitude compared to the baseline. The main Leader effect absorbs the pandemic premium and gets closer to the results without the COVID interactions. To further isolate the impact of the original COVID period on the analysis, column (3) of Table 6 restricts the sample to the pre-pandemic period, still resulting in negligible and insignificant placebo pandemic premium.

The second, and more substantial, challenge to identification relates to confounding as the assignment to leader regions is not a strictly randomized process. In the Empirical Strategy section, we noted the potential repercussions of systematic differences between regions, that could in turn be differentially impacted by pandemic related shocks. In particular, we recognize population density (strongly associated with urbanization levels) as potentially a critical confounder, and a helpful avenue to illustrate the ideas. As suggested before, if the pandemic has a differential impact on high- vs. low-density areas in terms of NTLs, our estimates would mistakenly attribute a density premium to regional favoritism.

Table 6: The COVID-19 Premium for Leader Birthplaces

	(1)	(2)	(3)	(4)	(5)
	Baseline	Placebo (full sample)	Placebo (pre-2020)	Density Deciles	Density Quartiles
Leader_{ict}	0.025* (0.013)	0.046*** (0.013)	0.025* (0.013)	0.038*** (0.013)	0.034*** (0.012)
$\text{Leader}_{ict} \times \text{COVID}_t$	0.133*** (0.020)			0.052*** (0.018)	0.069*** (0.017)
$\text{Leader}_{ict} \times \text{Placebo COVID}_t$		-0.014 (0.016)	0.018 (0.013)		
Observations (millions)	7.25	7.25	4.44	7.25	7.25
Regions	47,517	47,517	47,517	47,517	47,517
Country-Month FE	Yes	Yes	Yes	No	No
Ctry-Month-Density FE	No	No	No	Yes	Yes
Interaction Window	COVID (2020-21)	Placebo (2016-17)	Placebo (2016-17)	COVID (2020-21)	COVID (2020-21)

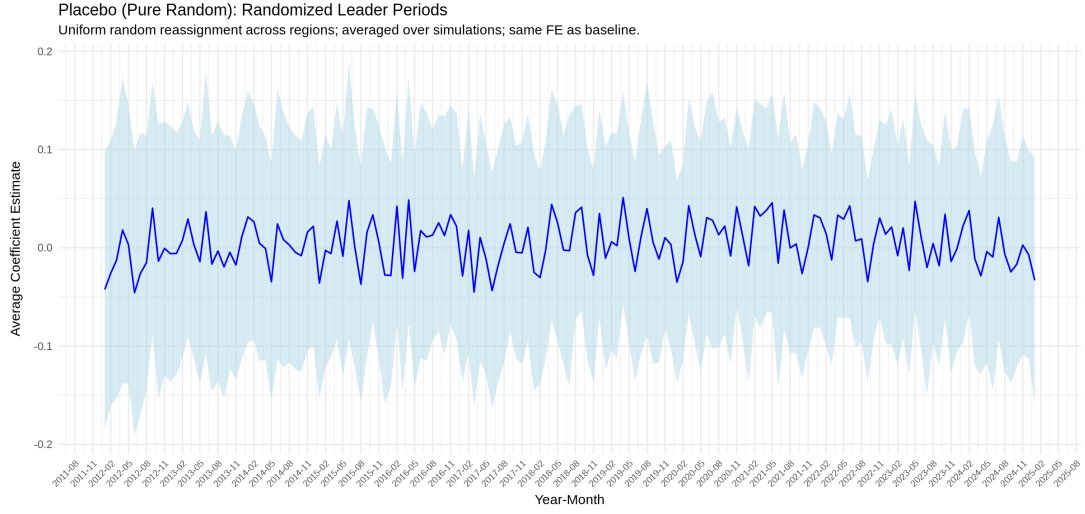
Notes: COVID window is Jan 2020–Dec 2021. Column (2) uses a placebo window of Jan 2016–Dec 2017 in the full monthly sample. Column (3) restricts the sample to years ≤ 2019 and applies the placebo COVID window. “Density Deciles/Quartiles” add country $\times$ month $\times$ population-density bin fixed effects (bins pre-defined using pre-2020 GPW). All models include region fixed effects; standard errors (in parentheses) are clustered by leader-term (country-month cells sharing the same set of leaders). Significance: * 10%, ** 5%, *** 1%.

We probe whether the spike in the pandemic premium can be generated as an artifact of the systematic differences between leader regions and others by means of two placebo simulations. As a sanity check, we first randomly reassign leader tenures in a uniform manner to different regions within the same country while retaining the original timing. Next, we re-estimate 5 for the simulated panel. We expect the resulting coefficients to have a calendar time profile hovering around zero. This conjecture is validated in Figure 3a. The resulting trajectory is a noisy path centered around zero.

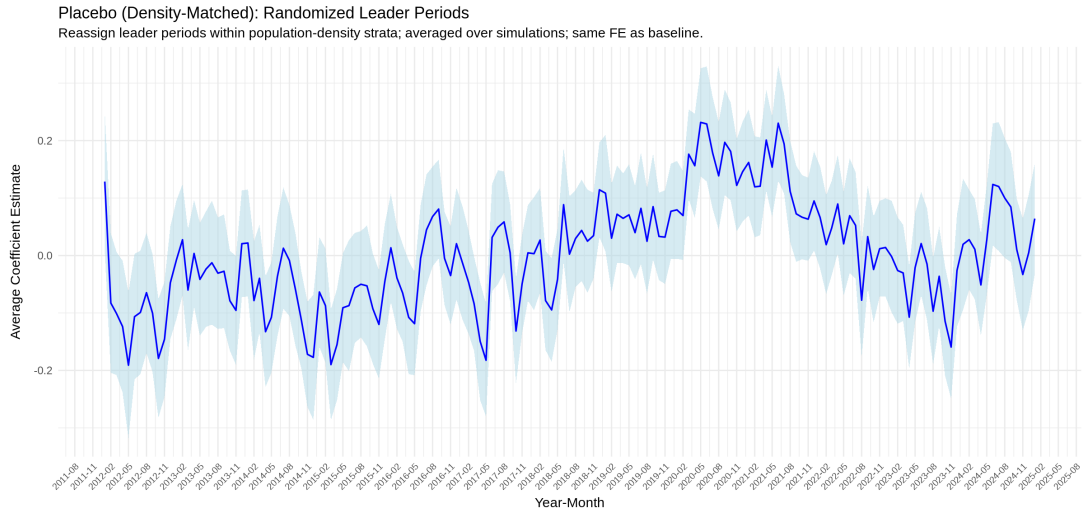
Next, we introduce a simulation exercise where we reassign placebo leader tenures with probabilities proportional to regional population density within a country.²⁷ The dynamic coefficient averages arising from these simulations are plotted in Figure 3b. The resulting image is remarkable. The trajectory of this placebo birthplace premium follows a similar trajectory to that of the actual leader data. This includes a strong COVID-timed bump. Existence of this pattern, especially the similarities during the pandemic, suggests that differential shocks by density are an important confound and validates our concern regarding not interpreting the coefficient on the interaction term $\text{Leader}_{ict} \times \text{COVID}_t$ purely as an indicator of favoritism.

The next section addresses concerns regarding the COVID-bump and proposes two methods to tackle them. First, we recognize population density as *the* major confounder and directly control for it using high dimensional fixed effects. Our second approach constrains the control set to a set of regions plausibly similar to the actual leader regions, namely to those regions that have been leader regions in the past.

²⁷For each leader tenure, the original leader during that tenure is excluded from the pool of eligible placebo leaders for the purposes of this exercise.



(a) Placebo: uniform random assignment within country



(b) Placebo: density-weighted assignment within country

Figure 3: Placebo simulations of leader–month dynamics under randomized leader assignments.

Notes: Each panel plots the average monthly coefficients from 50 simulation draws. Confidence bands around the coefficients corresponds to simple averaging across 95% confidence intervals for each simulation, each coefficient. In (a), leader status is reassigned uniformly at random across regions within a country, holding tenure timing fixed. In (b), reassignment probabilities are proportional to baseline population density.

5.4.2 Strategy 1: High-Dimensional Fixed Effects

Our first approach to resolve the identification issues surrounding the pandemic related leader birthplace premia is to acknowledge population density as the main confounder and directly control for it using a set of high-dimensional fixed effects in the form of *country-month-density bins*. This approach purges our estimates of shocks and trends across population density lines and ensures that leader birthplaces are compared only to other regions in the same country, during the same month, and within the same population density bin. The use of population density bin-linked fixed effects to control for time-varying heterogeneity related to densities has precedent in recent literature such as Kuchler et al. (2020); Coven et al. (2023).

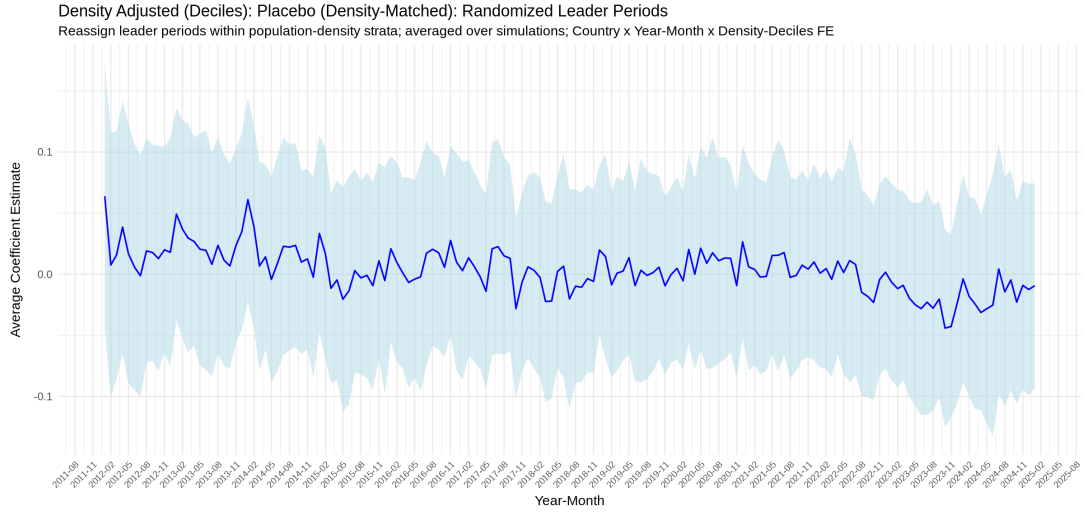
We operationalize these fixed effect interactions as represented by $\beta_{ct \times d(i)}$ in Eq. 6. Here, $d(i)$ corresponds to the density bin, where our preferred approach²⁸ is to define bins as population density deciles which maintains a considerable level of control for density related shocks while retaining sufficient variation. To calculate country specific density deciles, we rely on the Gridded Population of the World project from the Center for International Earth Science Information Network. In particular, we fix densities calculated from the 2015 version of this project.

The results from this specification are presented in Columns (4) and (5) of Table 6. Our estimate of the pandemic premium is 0.052 (0.069) for population density decile (quartile) interactions. This corresponds to a reduction compared to the baseline estimate of 0.133 from column (1) without the interactions. Crucially, our main finding of this section is that the pandemic premium after controlling for population densities is still positive, economically meaningful ($\geq 5\%$), and highly statistically significant. The boost in the coefficient of the base leader premium ($0.025 \rightarrow 0.038$) is also indicative of part of the leader premium being attributed to the COVID interaction $\text{Leader} \times \text{COVID}$ when population densities are not accounted for.

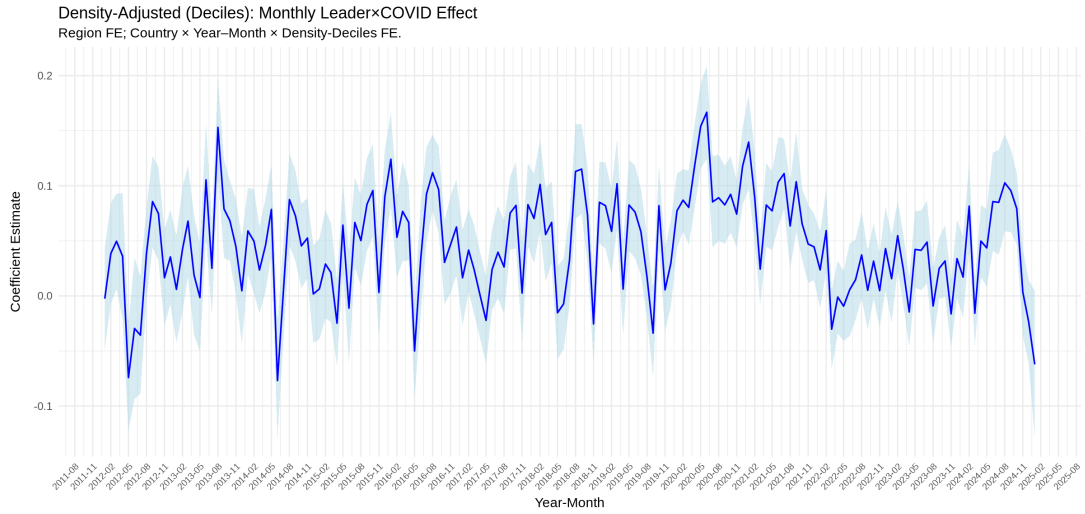
Finally, we apply the density-bin framework to the calendar-time diagnostic in Eq. 5 by replacing β_{ct} with $\beta_{ct \times d(i)}$ to study the impact on the temporal trajectory. To this end, we first validate this design by applying it to the density-weighted placebo leader simulations. Figure 4a shows that the spurious COVID bump completely disappears once country-month-density fixed effects are included, demonstrating that the proposed framework is effective at absorbing shocks that may originate purely from the capability of regions to produce leaders based on their population density characteristics. More critically, applying the same procedure to the actual data, Figure 4b presents a noticeably less noisy path outside of the COVID period and a softened yet clear and precisely estimated elevation during pandemic’s most impactful period.

The observations listed above point out the importance of controlling for regional features related to population densities. As discussed in the empirical strategy section, country-month-density bin fixed effects may not necessarily capture an exhaustive set of factors that might confound the analysis. The next subsection suggests an alternative approach which focuses on sample design rather than explicit controls.

²⁸We will use population density deciles for the rest of the analysis unless specified otherwise.



(a) Placebo with country–month–density FE (deciles)



(b) Actual data with country–month–density FE (deciles)

Figure 4: Density-adjusted dynamics of the leader effect (monthly).

Notes: Both panels plot monthly coefficients on the leader indicator from models that include region fixed effects and country–month–density decile fixed effects (density bins calculated from GPWv4 2015 raster). Panel (a) uses density-weighted placebo leader assignments; Panel (b) uses actual leader terms. In Panel (a), confidence bands around the coefficients corresponds to simple averaging across 95% confidence intervals for each simulation, each coefficient. In Panel (b) vertical bars show 95% confidence intervals.

5.4.3 Isolating the COVID Premium: A Past-Leaders Control Group

Our analysis above recognized population density as an important observed heterogeneity and established a significant pandemic premium even after controlling for shocks and trends surrounding this confounder. However, leader-producing regions might be different from other regions in unobservable dimensions such as political centrality, industrial structure, demographic and ethnic characteristics etc. which might in turn be correlated with economic outcomes during a crisis, potentially resulting in biased estimates. We address this concern by providing a COVID-anchored event-study that focuses only on regions that have produced national leaders. We restrict attention to Past Leaders as a counterfactual to actual COVID Leaders because these regions have demonstrated the capacity to produce national leaders and they are likely to share unobserved characteristics with those of COVID Leaders.

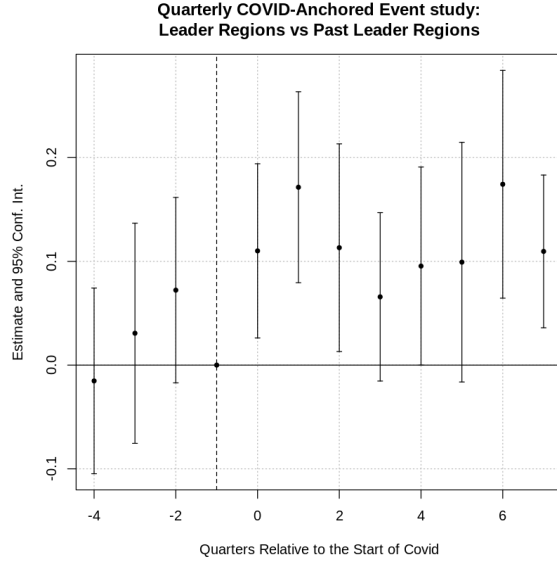
Eq. 7, which is detailed in empirical strategy section, informs the analysis in this section. This specification traces the evolution of the gap in NTLs between treated and control regions, where the gap is normalized to zero in quarter -1 (and month -1 for monthly estimates), the quarter (month) right before the start of the pandemic, which is fixed at January 2020. We require treated regions to retain the leader status during 1 year before and 2 years after the pandemic’s start. While this reduces the number of countries that could be included, it also provides cleaner episodes of NTL trajectories as our interest is to observe these dynamics further into the pandemic. In a similar vein, we require Past Leaders to have substantial leadership tenures (at least 12 months) where their terms end before 2016, and do not repeat thereafter.

As a benchmark, we first implement a direct analogue of the above framework regarding the restricted leader producing regions sample to an expanded sample of all regions, while still requiring COVID Leaders to maintain leader status within the window described above. This is a refinement of the analysis for the calendar-time dynamics of the birthplace premium plotted in Figure 2. The results, deferred to Figure 8 in Appendix D, mirrors the COVID spike seen in Figure 2.

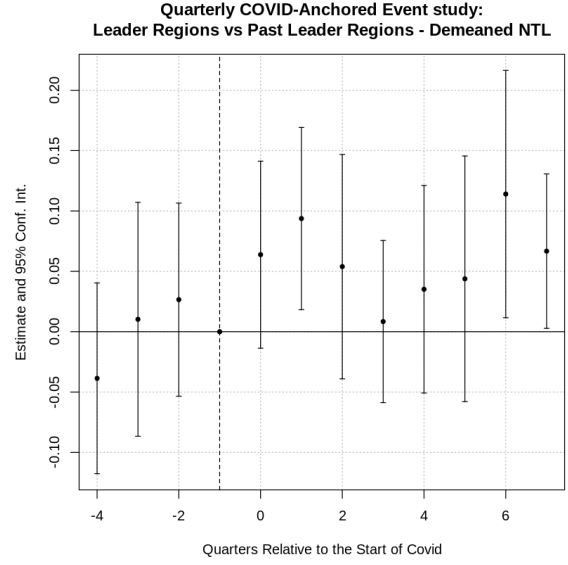
Moving on to the leaders sample framework outlined above, Figure 5 presents the results of the event study. Panel (a) shows the quarterly estimates of the raw logged NTL gap between COVID Leaders and Past Leaders. The pre-period coefficients are clustered around zero and insignificant, confirming that these two groups were evolving on similar trajectories in terms of their nighttime luminosity. A positive and significant gap emerges by the start of the pandemic. The effect remains elevated around 10% throughout the pandemic. In Panel (c), we apply the same model to monthly data. The pre-trend clusters around zero are more clearly visualized. Moreover, the pandemic bump is showing similar patterns except that the coefficient estimates are less precise, predominantly due to higher volatility of luminosity readings in shorter time horizons.

Panels (b) and (d) of Figure 5 incorporates the idea from the previous strategy, i.e. controlling for population density related shocks. Here, we create a new outcome variable “Demeaned NTL” that subtracts the country-month-density bin mean (quarter in this case) from the raw NTL after log transformation. The trajectory of the premium gap is similar to the case without demeaning, albeit less precise and moderated to around 5-6%.

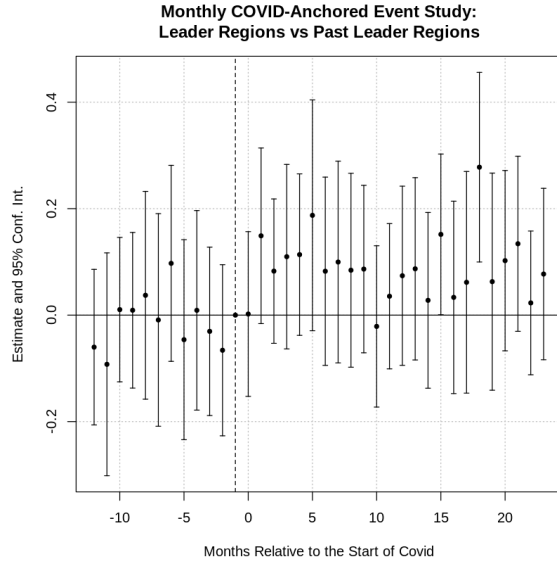
We extend the idea of restricting attention to “politically relevant regions” by incorporating



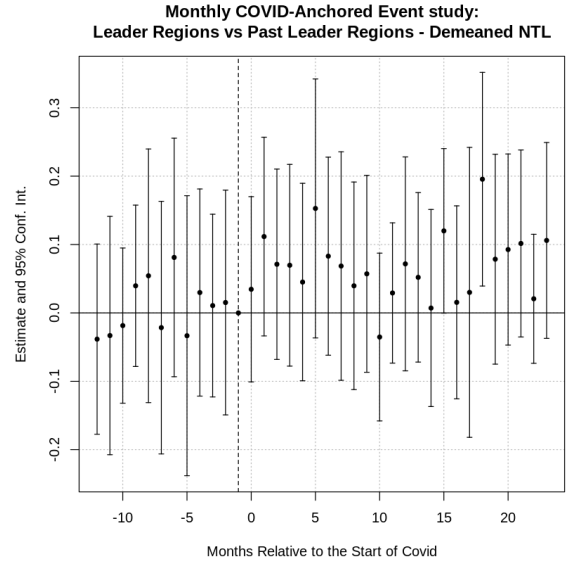
(a) Quarterly, Raw NTL



(b) Quarterly, Demeaned NTL



(c) Monthly, Raw NTL



(d) Monthly, Demeaned NTL

Figure 5: COVID-anchored event study comparing “COVID Leaders” to “Past Leaders”.

Notes: Each panel plots coefficients from Eq. 7 event study anchored at January 2020 ($\tau = 0$). Treated regions (“Leader Regions”) are those with a continuous leader term spanning at least 12 months before and 23 months after the anchor; controls (“Past Leader Regions”) are regions that had a ≥ 12 -month leader term ending by Dec 2015 and no leader status after Dec 2015. The estimating equation includes region fixed effects and country-event time τ fixed effects; the baseline period is $\tau = -1$. The outcome is $\log(NTL + 0.01)$. “Demeaned NTL” subtracts the country-month-density bin mean (density bins from GPW-2015, by country; quartiles). Monthly panels plot month-level estimates; quarterly panels average months within quarters before estimation. Error bars show 95% CIs with standard errors clustered at the country level.

Table 7: COVID-19 Premium within Politically Relevant Regions

	(1) Average Effect	(2) COVID DiD
Leader _{ict}	0.043*** (0.013)	0.034** (0.014)
Leader _{ict} × COVID _t		0.066*** (0.022)
Sample	—— Politically Relevant Regions ——	
Observations	126,693	126,693
Regions	834	834
Region FE	Yes	Yes
Country–Month FE	Yes	Yes

Notes: Dependent variable is $\log(\text{NTL}_{ict} + 0.01)$ where NTL is calculated from the monthly near-nadir composites. “Politically Relevant Regions” are ADM2 units that have ever produced a national leader. COVID window: Jan 2020–Dec 2021. Standard errors (in parentheses) clustered by leader term (country–month cells sharing the same set of leaders). Significance: * 10%, ** 5%, *** 1%.

it into our main specification in Eq. 3. In this setup, the sample only contains regions that have ever produced a national leader. A total of 834 regions, corresponding to approximately 2% of the total number of regions, satisfy this condition. Table 7 presents the results from this analysis. Column (1) of the table estimates the average leader premium for the 2012–2025 period among politically relevant regions. The coefficient 0.043 ($\approx 4.4\%$) closely follows the unrestricted analysis. Column (2) adds the COVID interaction and measures the coefficient at 0.066, corresponding to a pandemic premium of $\approx 7\%$ within this leader producing regions.

The findings from both approaches to constraining the sample to only leader producing regions exhibit positive and significant pandemic premia at similar magnitudes. Moreover, a comparison of this premium with the population density factored fixed effects approach also yield commensurate estimates. These two complementary analyses provide our most credible evidence of a genuine crisis premium in the context of COVID-19 pandemic.

5.4.4 Validating the Empirical Strategy: A Placebo Simulation

So far, we presented population density factored high-dimensional fixed effects structure as the preferred framework for the rest of the analysis. Even though the two-pronged empirical approach points to the same conclusion regarding the pandemic premium, the quantitative findings are not directly comparable. Therefore, a crucial question still remains: how effective is the population density approach in eliminating confounding factors. To this end, we implement a falsification exercise that builds a placebo treatment which replicates the space of confounders that threatens identification, but has no real causal impact.

We operationalize the falsification exercise via simulations that reassign actual leader terms to other regions that had produced a leader in the past within the same country. For each placebo dataset we estimate Eq. 3 (i) with country-month fixed effects, and (ii) country-month-density decile fixed effects. By construction, the placebo leader regions share leader-like characteristics with the actual leader regions and we expect them to exhibit spurious pandemic bumps that are due to the interaction of these characteristics and the NTL emissions. Therefore, if the

population density fixed effects approach works as what we intend it to, the spurious COVID bump in (i) should be mitigated, and ideally collapse to zero in (ii).

Table 8: Placebo Simulations for the COVID Premium: Effect of Density Fixed Effects

	(1) Without Density	(2) With Density
Mean placebo estimate	0.066	0.020
Median placebo estimate	0.067	0.020
Mean std. dev. of estimates	0.018	0.017
Share of simulations with $p < 0.05$	96%	14%
Share of simulations with positive estimate	100%	94%
Number of simulations	50	

Notes: Each column summarizes the distribution of the estimated coefficient on the placebo interaction $\text{Leader}_{ict} \times \text{COVID}_t$ in Eq. 3 across 50 simulations. In every simulation, leader status during the COVID window (Jan 2020–Dec 2021) is reassigned to a random region within the same country that produced a leader prior to 2012. Column (2) augments the baseline with country $\times$ month $\times$ population-density decile fixed effects (deciles computed from GPWv4 2015, by country).

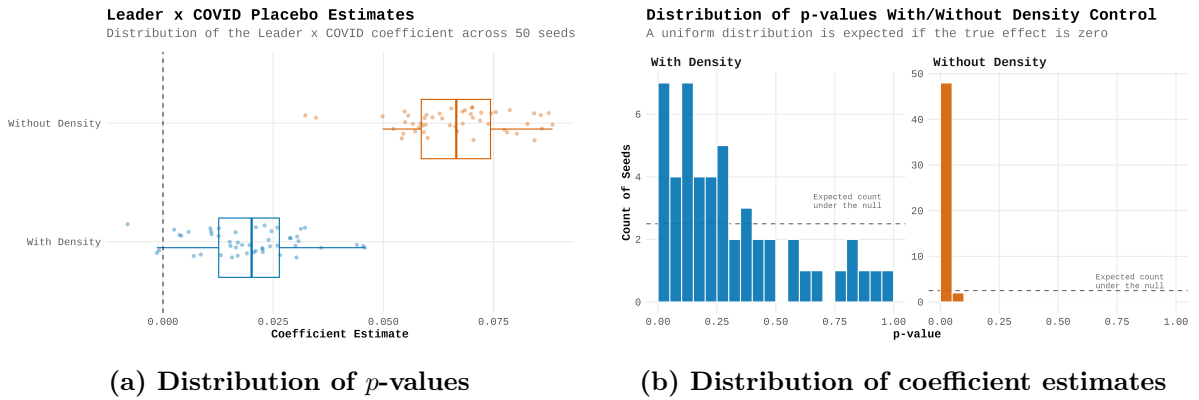


Figure 6: Validating the density-adjusted fixed-effects strategy with placebo simulations

Notes: Results from 50 simulations where leader status during the COVID window (Jan 2020–Dec 2021) is reassigned to “past-leader” regions within the same country. The *With Density* model includes region and country–month–population-density decile fixed effects (deciles based on GPWv4 2015, computed by country). The *Without Density* model uses country–month fixed effects instead. Panel (a) shows the distribution of the coefficient estimates on $\text{Leader} \times \text{COVID}$. Panel (b) shows the histogram of p -values for the placebo COVID interaction;

Figure 6 and Table 8 summarize the results across 50 simulations. The *Without Density* model that doesn’t control for population density consistently finds a large ($\approx 7\%$) placebo premium. Moreover, Panel (b) of Figure 6 shows that these estimates are significant at 0.05 levels in 96% of the simulations. Considering the expected rate of 5% rejections under the null, this exercise successfully diagnoses the problem: a spurious pandemic premium is generated due to characteristics that make a region a *leader region*.

The model with the population density controlled via country $\times$ month $\times$ density decile fixed effects substantially mitigates the spurious pandemic premium problem. Average pandemic premium estimates drastically drop to 2%. Moreover, only 14% of these estimates are statistically significant at 5% levels, and the graph of p -values in Figure 6 show a fairly flat profile, with a relatively small bias towards failing to reject the spurious premium hypothesis.

This exercise provide support towards validating our empirical strategy. It shows that there are real confounding characteristics related to leader producing regions that yields biased pandemic premium estimates in simpler models. More crucially, it demonstrates that the population density bin adjusted fixed effects structure is capable of absorbing the bulk of the bias introduced by the aforementioned characteristics, lending credibility to parts of our analysis that use this approach.

5.5 The Moderating Role of Democratic Institutions

Table 9: Democratic Institutions and the COVID Premium in Birthplace Favoritism

	(1) Full Sample Avg. Democracy Effect	(2) Full Sample Full Interaction	(3) High-Dem. Subsample	(4) Low-Dem. Subsample
Leader	0.072*** (0.023)	0.060** (0.023)	0.016** (0.008)	0.088* (0.037)
Leader $\times$ High-Democracy	-0.055** (0.024)	-0.043* (0.024)		
Leader $\times$ COVID		0.083*** (0.028)	0.003 (0.015)	0.080** (0.028)
Leader $\times$ COVID $\times$ High-Democracy		-0.079** (0.031)		
Country-Month-Density FE	Yes	Yes	Yes	Yes
Observations	7,254,532	7,254,532	4,351,976	1,881,038
Regions	47,517	47,517	28,369	12,509

Notes: Dependent variable is $\log(\text{NTL}_{ict} + 0.01)$ (VIIRS, monthly near-nadir). “High(Low)-Democracy” is a dummy for countries in the top (bottom) tercile of the 2018 Polity5 index. All models include region fixed effects and country-month-population-density decile fixed effects (deciles built from GPWv4 2015 by country). Standard errors (in parentheses) are clustered by leader period (country-month cells sharing the same set of leaders). COVID window: Jan 2020–Dec 2021. Significance: * 10%, ** 5%, *** 1%.

Does the quality of political institutions restrict the ability of a leader to divert resources to their birthplaces during a crisis? In this section, we test this hypothesis in the context of the COVID-19 pandemic with an emphasis on the role of the democratic checks and balances. Democratic quality is proxied by the widely adopted polity2 score from the Polity5 index where we classify countries into terciles based on their 2018 scores. The top tercile countries are defined as *High-Democracy* and the bottom tercile countries are considered *Low-Democracy*.²⁹

Table 9 demonstrates 4 different specifications to uncover the role of democratic quality. Column (1) of the table prefaces the rest of analysis by focusing on whether established democracies mitigate the *overall* birthplace favoritism regardless of the periods. The baseline leader coefficient of 0.072 is mitigated by about 3/4s by the High-Democracy countries, indicating an overall drop in birthplace favoritism. A Wald test on the sum of Leader + Leader $\times$ High-Democracy shows that the net effect for strong democracies is approximately 1.7% and significant at 5% level. This result is consistent with H&R where the authors noted that birthplace favoritism disappears at the top of the democratic quality range (with a polity score of 10). In contrast, we find residual

²⁹The top tercile in our sample corresponds to countries with a Polity score of +8 or higher.

average favoritism for a relatively wider set of countries.

In Column (2) we introduce the triple interaction term $\text{Leader} \times \text{COVID} + \text{Leader} \times \text{COVID} \times \text{High-Democracy}$ to investigate the main question of pandemic favoritism mitigation by High-Democracy group. The coefficient 0.083 ($\approx 8.7\%$) on the pandemic premium is virtually erased for the countries in the High-Democracy group. Columns (3) and (4) split the sample into High- and Low-Democracy subsamples. The findings paint a strongly consistent pattern. For High-Democracy subsample, the overall birthplace premium is quite modest at 1.6%, and there is no additional COVID premium. At the bottom of the democratic quality, the overall premium goes up to 9% and doubled during COVID.

Our analysis in this section provides a stark contrast between strong and weak democratic institutions in terms of curbing birthplace favoritism both within and outside COVID-19 period. Even though a small degree of regional favoritism remains for established democracies, the ability to exploit a crisis towards birthplace favoritism appears to be a feature of countries primarily with weaker institutional constraints.

6 Conclusion

In this paper, we revisited the question of whether national leaders systematically favor their birthplaces, a conclusion first documented at a global scale by Hodler and Raschky (2014). Our analysis first took on the challenge of replicating H&R’s main findings using publicly available datasets. Results from this investigation demonstrated overall replicability, with a divergence for contemporaneous estimates, which we understand, among others, is due to how partial leader terms would be defined. Such a discrepancy is directly inherited from annual frequency of DMSP-OLS data which doesn’t allow for a closer look.

More than a decade later, we have better remote sensing instruments, significantly improving temporal and spatial resolution, and incorporating detailed quality flags. We confirm regional favoritism in the contemporary era (2012-2024) with higher quality VIIRS NTL datasets. Not only a wide range of robustness exercises is possible and shown to hold birthplace premium, but also the temporal resolution allows for a granular analysis of leadership transitions. Monthly composites reveal that benefits proxied by NTLs accrue to the leader birthplaces as soon as their accession to power and remain in effect long after their departure, conclusions not apparent in previous analyses, pointing out the value of new generation remote sensing data for studying political economy questions.

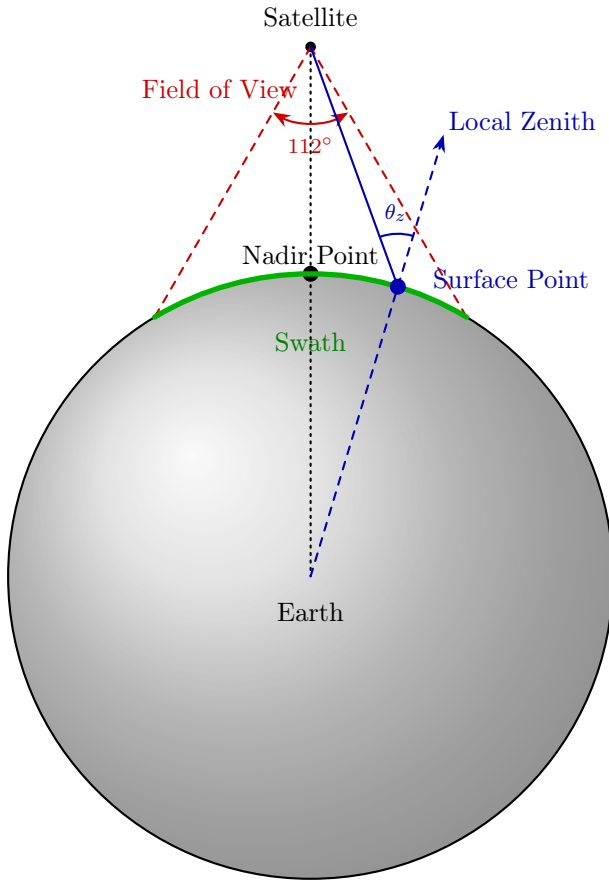
Our core novel contribution also stems from the temporal precision of the VIIRS NTLs where we investigate the pattern of birthplace favoritism during a crisis. We observe a striking COVID-19 pandemic bump in luminosity values for leader birthplaces, but a naive reading of such findings is problematic due to potential contamination from confounders resulting from systematic differences of these regions and the rest within the same country, particularly population density being an important factor, and subsequent interaction of NTLs with these confounders. We present two approaches, i.e. directly controlling for population density related shocks and a restricted control group of past leader regions, to isolate a meaningful pandemic premium. We also find that the crisis premium is a feature of countries with weaker democratic quality, suggesting

that checks and balances afforded by strong democracies are effective in constraining leaders' ability to divert resources in crises periods.

Our results provide a clear picture: better data allows us to conclude that birthplace favoritism is a thing of both the past and the present, and it can be exacerbated during periods of heightened discretionary spending. Moreover, democratic accountability is an important determinant in mitigating such concerns. A subject of future research is further refining the empirical strategy. To this end, regions that could have produced leaders but narrowly missed, i.e. birthplaces of the main political competitors of actual leaders, provide a well suited control group framework similar to the past leader regions. Also of interest, but beyond the scope of this paper, is the detection of specific mechanisms of the pandemic induced favoritism such as fiscal transfers and procurement contracts. Finally, COVID-19 pandemic is a very specific global crises and the applicability of the political favoritism to a broader range of disasters and crisis is an open question.

Appendices

A Remote Sensing Terminology



Field of View (FOV). The instrument’s field of view corresponds to the angular extent over which the sensor observes the Earth’s surface during a single scan. The VIIRS FOV is approximately 112° , and the satellite is located at an altitude of 829 km.

Zenith Angle θ_z . The local zenith angle is the angle between the line of sight from the satellite to a surface point and the line straight from the center of Earth (Local Zenith). It increases as the satellite moves away from nadir, from 0° at the nadir up to a maximum of 56° at the VIIRS scan edge.

Swath. The swath is the ground measured width of the scan. The full VIIRS swath is approximately 3040 km.

B List of Additional Countries and Territories

The following is the list of countries and territories used in the global analysis and not covered by H&R:

Åland, United Arab Emirates, Armenia, American Samoa, Azerbaijan, Bahrain, Bahamas, Saint-Barthélemy, Belize, Barbados, Brunei, Switzerland, Comoros, Cabo Verde, Cuba, Cyprus, Djibouti, Dominican Republic, Egypt, Estonia, Faroe Islands, Micronesia, Guadeloupe, Equatorial Guinea, French Guiana, Croatia, Hungary, Ireland, Iceland, Israel, Jamaica, Kyrgyzstan, Kuwait,

Libya, Lesotho, Luxembourg, Moldova, Malta, Martinique, Mauritius, New Caledonia, Palestine, Qatar, Réunion, Romania, Saudi Arabia, Singapore, Saint Helena, Ascension and Tristan, Solomon Islands, South Sudan, São Tomé and Príncipe, Suriname, Swaziland, Syria, Turkmenistan, Tonga, Trinidad and Tobago, Turkey, Uzbekistan, Virgin Islands (U.S.), Vanuatu, Wallis and Futuna, Samoa, Kosovo, Northern Cyprus.

C VIIRS–DMSP Comparison (2012–2013)

Table 10 compares DMSP-OLS and VIIRS datasets for 2012–2013 (the overlapping years).

Table 10: Mapping VIIRS to DMSP-OLS, 2012–2013 (log–log)

Spec	Sample	FE	Clust.	$\hat{\beta}$ on $\ln(\text{DMSP})$	N (Within R^2)
(1)	All	Ctry×Yr	ADM2	0.9703 (0.0039)***	93,294 (0.762)
(2)	All	Yr only	ADM2	0.9832 (0.0026)***	93,295 (0.826)
(3)	All	Ctry×Yr	Country	0.9703 (0.0303)***	93,294 (0.762)
(4)	DMSP > 0	Ctry×Yr	ADM2	1.1110 (0.0038)***	86,348 (0.817)
(5)	DMSP > 0, DMSP < 55	Ctry×Yr	ADM2	1.0522 (0.0038)***	82,712 (0.809)
(6)	DMSP > 0, DMSP ≥ 55 interact.	Ctry×Yr	ADM2	see notes	86,348 (0.841)

Notes: Dependent variable is $\ln(\text{VIIRS} + 0.01)$. Standard errors clustered as shown. All $p < 0.01$. Specification (6):

$$\ln(\text{VIIRS} + 0.01) = 1.0535 \ln(\text{DMSP} + 0.01) - 32.5166(\text{DMSP} \geq 55) + 8.3043 \ln(\text{DMSP} + 0.01) \times (\text{DMSP} \geq 55).$$

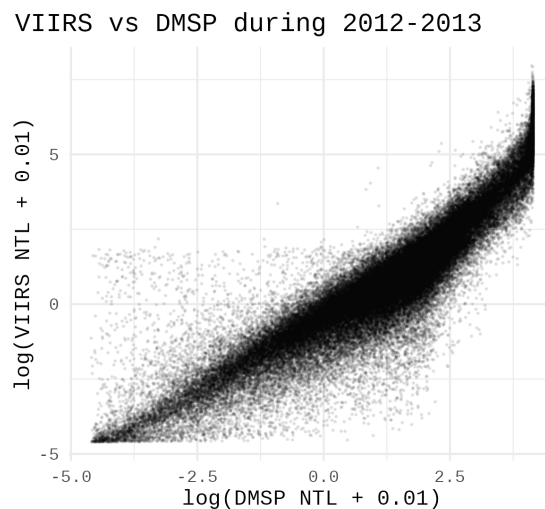
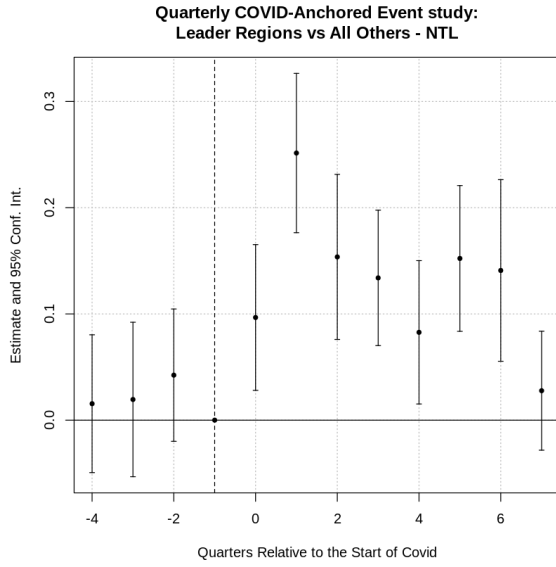
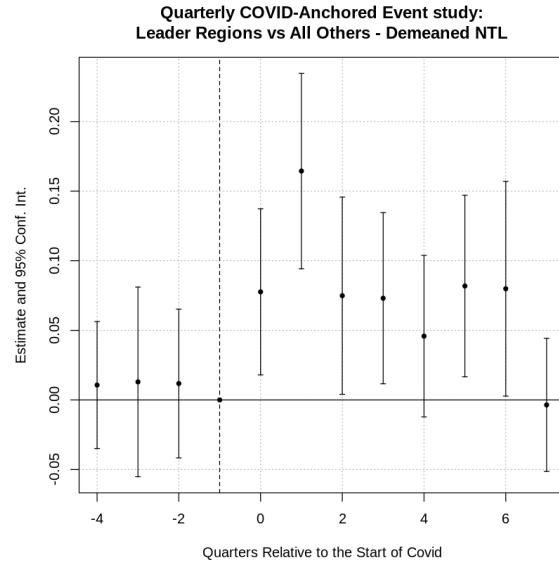


Figure 7: VIIRS vs DMSP, 2012–2013.

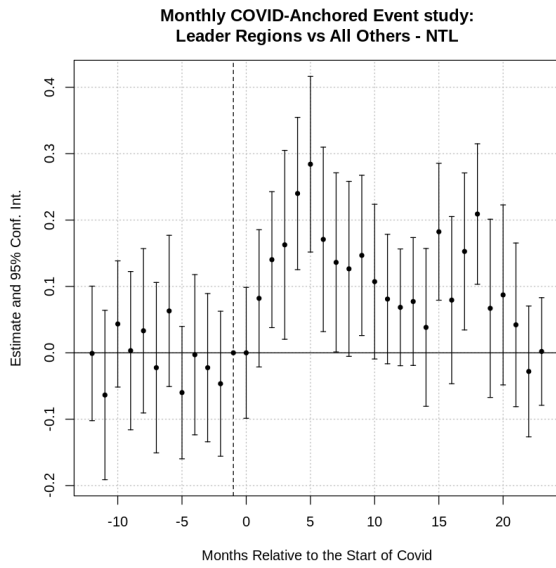
D COVID-Anchored Event Study for All Regions



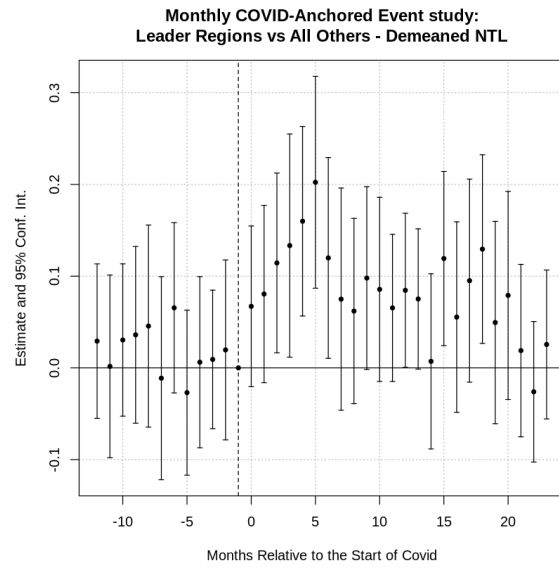
(a) Quarterly, Raw NTL



(b) Quarterly, Demeaned NTL



(c) Monthly, Raw NTL



(d) Monthly, Demeaned NTL

Figure 8: COVID-anchored event study comparing regions with a leader during the pandemic to all other regions.

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